HEAVENSTAY WELLNESS RESORT MANAGEMENT SYSTEM

Web UI Demo — Functional & UI/UX Overview

A Comprehensive Technical & Architectural Synthesis of the 56-Screen Front-End Prototype, Domain Workflows, Role-Based Access Boundaries, and Verification Baselines.

PROJECT NAME
HSWRMS (HeavenStay Wellness Resort Management System)

COURSE & ACADEMIC CONTEXT
ISP392 — IS2005 Group 1 (Fall 2026 / Spring 2025)

MODULE SCOPE
M1 through M6 (Full 6-Module Domain Lifecycle)

SCREEN SCOPE BASELINE
Official 56 Baseline Screens (100% Implemented in Prototype)

FRONTEND STACK BASELINE
React 19, TypeScript, Vite 8, React Router v7, Vanilla CSS Tokens

PRIMARY DOCUMENT PURPOSE
ISP392 Defense, Demo Walkthrough, SRS Cross-Check & Onboarding

Classification: Academic Project Specification & System Review
Document Revision: 2.0 (Official Baseline Alignment)
Date: October 2026

Table of Contents

1. Executive Overview 3
2. UI/UX Design System Overview 5
3. Role-Based UI Architecture 8
4. Master Screen & Navigation Overview 11
5. Module 1 — Authentication & Sensitive Health Profile 15
6. Module 2 — Retreat Booking & Accommodation 18
7. Module 3 — Spa & Therapy Scheduling 21
8. Module 4 — Dietary F&B Management 26
9. Module 5 — Consolidated Checkout & Analytics 30
10. Module 6 — System Administration & Permission Management 34
11. Screen-by-Screen UI Catalog (Screens 1 to 56) 38
12. Status & State Reference Guide 46
13. Cross-Module UI Dependencies & Data Flow 49
14. Prototype vs Approved Business Requirements (Gap Analysis) 51
15. UI/UX Quality & Usability Review 54
16. Demo Walkthrough Script (10–15 Minutes Presentation) 57
17. Mock Viva: 20+ Challenging Defense Questions & Answers 60
18. Open Architectural Items & Known Limitations 66
19. Technical Frontend Summary & Source Topology 69
20. Final Assessment & Defense Readiness Checklist 71

Document Scope & Methodological Integrity Notice

This document constitutes the definitive architectural and functional review of the **HeavenStay Wellness Resort Management System (HSWRMS)** front-end prototype. All statements, component inventories, route mappings, and state transitions described herein have been cross-verified against the actual React 19 source code, the official `Screen_List.md` inventory, and the authoritative Software Requirements Specification (`HSWRMS_ISP392_G1_SRS.pdf`). Where current front-end prototype behaviors diverge from approved SRS business rules, both states are documented explicitly in the Gap Analysis without silent reconciliation.

1. Executive Overview

1.1 What HSWRMS Is

HeavenStay Wellness Resort Management System (HSWRMS) is an integrated enterprise hospitality management platform purpose-built for luxury wellness resorts. Unlike conventional Property Management Systems (PMS) that focus narrowly on room inventory and basic folio billing, HSWRMS coordinates multi-day holistic wellness retreats combining sanctuary accommodations, personalized therapeutic spa sessions, tailored nutritional culinary programs, and statutory residence compliance into a cohesive guest experience.

1.2 Why the Web UI Demo Exists

The front-end Web UI Demo was created to serve as an interactive, high-fidelity presentation baseline for the **ISP392 Software Project**. It provides complete visual, structural, and behavioral verification of all user journeys prior to full production backend deployment. By implementing the entire 56-screen inventory with interactive state management, simulated gateway workflows, and role-adapted layouts, the demo enables academic examiners, stakeholders, and project members to evaluate system completeness, role boundaries, sensitive data isolation, and user experience coherence.

1.3 Target Users Supported

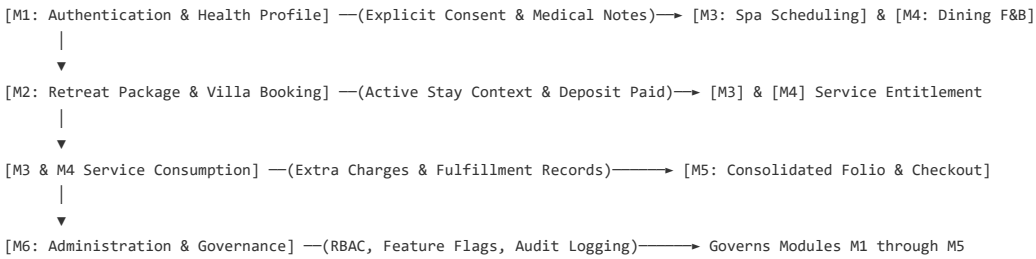
HSWRMS supports six distinct authenticated operational personas alongside unauthenticated public visitors:

- **Public Visitor (Unauthenticated):** Explores retreat packages, villa suites, resort philosophies, and initiates account registration or authentication.
- **Customer:** The resort guest (strictly termed *Customer*, never *Guest* in authenticated contexts). Manages confidential wellness profiles, reserves multi-day retreats, schedules included and extra therapies, pre-selects dietary meals, monitors in-house folios, and submits post-checkout feedback.
- **Receptionist (Front Desk):** Manages daily arrivals, executes legal identity verification and residence reporting (Vietnam Law on Residence 2020), assigns physical sanctuary villas, books manual extra therapies, performs billing audits, and finalizes resort departure.
- **Spa Therapist / Yoga Instructor:** Reviews assigned daily treatment schedules, accesses strictly treatment-relevant health precautions, and updates session fulfillment statuses (Completed or No-Show).
- **F&B Staff / Chef:** Manages culinary kitchen prep lists, receives allergen and dietary alerts while keeping sensitive medical records masked, and transitions meal orders from Preparing to Ready.
- **Manager:** Operational overseer operating under least-privilege principles (not an administrator). Monitors aggregated revenue, villa occupancy, therapist utilization, and system-wide operational health.
- **Admin (System Administrator):** Configures staff credentials, maintains master catalogs, tunes business parameters, toggles feature flags, enforces the RBAC permission matrix, delegates limited operational rights, and audits sensitive data access.

1.4 How the UI Connects the Six Modules

HSWRMS is architecturally partitioned into six tightly coupled functional modules, linked by an end-to-end data chain:

End-to-End Enterprise Module Chain



1.5 Core Distinction: Conventional PMS vs. Wellness Resort Management System

DIMENSION	CONVENTIONAL HOTEL/RESORT PMS	HEAVENSTAY WELLNESS RESORT (HSWRMS)
Primary Unit of Sale	Physical room night (Room Only or Room + Breakfast).	Comprehensive Wellness Retreat Package combining lodging, daily therapies, and nutritional meal plans.
Guest Health Data	Usually uncollected or limited to informal text notes in guest comments.	Legally safeguarded Sensitive Health Profile requiring explicit consent under Vietnam PDPD decree; subject to post-stay erasure.
Data Minimization	Staff members typically share unrestricted view of guest reservation profile.	Role-adapted data projections: Therapist sees only muscular/joint precautions; Chef sees only food allergens; Receptionist sees no medical notes.
Spa Scheduling	Loose calendar booking; often detached from room stay or therapist availability.	Strict 2D Atomic Lock: Session confirmed only when both qualified practitioner and compatible treatment room are simultaneously available.
Dining Operations	POS ticket printing; kitchen unaware of customer medical restrictions.	Personalized culinary filtering based on dietary profile; ingredient-level requirement calculation and allergen cross-check.
Checkout Lifecycle	Payment of bill immediately completes checkout.	Decoupled two-stage process: Payment Success != Checkout Complete. Departure requires settling billing guards and reassigning villa to housekeeping.

2. UI/UX Design System Overview

2.1 Visual Philosophy: Quiet Luxury & Restrained Minimalist Hospitality

The visual interface of HSWRMS is implemented strictly in accordance with `HSWRMS_DESIGN.md`. Inspired by ultra-luxury wellness havens such as Amanoi Ninh Thuan, the design language rejects loud commercial gradients, heavy drop shadows, and high-contrast neon styling. Instead, it employs organic earth tones, soft stone surfaces, generous whitespace, tactile card elevations, and editorial serif typography.

2.2 Design Tokens & Visual Hierarchy

Color Palette & Surface Tones

- **Canvas Background:** `#FAF9F6` (Warm off-white stone canvas)
- **Soft Background:** `#F5F3EF` (Subtle container fill)
- **Card Surface:** `#FFFFFF` (Clean white surface elevation)
- **Primary Text:** `#2B2B2B` (Deep charcoal, never pure black)
- **Secondary Text:** `#706E6B` (Muted slate gray for subtitles)
- **Brand Forest Accent:** `#4F5A50` (Deep pine/forest green)
- **Brand Olive Accent:** `#6D7466` (Earthy natural olive)
- **Brand Sand Accent:** `#D8C7AE` (Warm coastal sand accent)

Typography & Shape Language

- **Display Typography:** `Cormorant Garamond` (Serif, elegant, applied to page titles, package headers, and editorial callouts).
- **UI & Tabular Typography:** `Inter` (Sans-serif, neutral, high legibility for data grids, badges, inputs, and numbers).
- **Radius Scale:** `6px` (inputs/chips), `10px` (controls), `14px` (buttons), `18px` (standard cards), `22px` (hero panels), `9999px` (pills).
- **Elevation & Shadows:** Tactile micro-shadows (`0 4px 20px rgba(30, 30, 30, 0.06)`) preventing visual noise.

2.3 Actor Badge Color Conventions

To ensure instantaneous cognitive recognition across complex cross-role operations, each actor role is assigned a distinct, low-saturation pastel identification token (matching `Screen_List.md §8`):

ACTOR PERSONA	BACKGROUND COLOR	HEX CODE	TEXT COLOR	OPERATIONAL APPLICATION
Public / Shared	Pastel Slate	`#E0E0E0`	`#424242`	Public landing, authentication modals, shared system screens
Customer	Soft Meadow Green	`#C8E6C9`	`#204B23`	Customer portal, booking cards, health profile indicators
Receptionist	Warm Primrose Yellow	`#FFF3B0`	`#665A00`	Front desk workspace, daily arrivals, villa check-in controls
Spa Therapist	Lavender Mist	`#D1C4E9`	`#3C2466`	Therapist schedule, session details, treatment notes
F&B Staff / Chef	Peach Amber	`#FFD8A8`	`#703C00`	Kitchen preparation board, dining orders, dietary warning cards
Manager	Quiet Seafoam Teal	`#B2DFDB`	`#004D40`	Revenue reports, occupancy analytics, therapist utilization
Admin	Sky Horizon Blue	`#BBDEFB`	`#0D47A1`	System dashboard, staff lists, feature flags, audit logs

2.4 Reusable Component Library

The frontend prototype avoids ad-hoc styling by constructing every screen from a dedicated library of typed, modular UI components located in `src/components/`:

COMPONENT NAME	SOURCE LOCATION	PRIMARY VARIANTS	UI CONSISTENCY BENEFIT
`Button`	`components/common/Button.tsx`	`primary`, `secondary`, `ghost`, `danger`	Standardizes quiet-luxury interactive states, loading spinners, and focus rings.
`Card`	`components/common/Card.tsx`	`default`, `elevated`, `interactive`, `bordered`	Ensures consistent 18px border radius, subtle borders, and soft padding across screens.
`StatusChip`	`components/common/StatusChip.tsx`	`success`, `warning`, `danger`, `neutral`	Harmonizes status indicators for booking, payment, and meal readiness.
`VillaStatusChip`	`components/operations/VillaStatusChip.tsx`	`available`, `occupied`, `maintenance`, `cleaning`	Encapsulates operational housekeeping statuses with color-coded dot badges.
`OperationalPageHeader`	`components/operations/OperationalPageHeader.tsx`	Operational with badge, date stamp, actions	Provides consistent top title hierarchy and screen number tag for all staff views.
`OperationalSidebar`	`components/navigation/OperationalSidebar.tsx`	Receptionist, Therapist, Chef navigation	Fixed left rail navigation featuring role badge, active indicators, and logout.
`CustomerNavbar`	`components/navigation/CustomerNavbar.tsx`	Customer navigation header	Includes brand logomark, stay shortcut, language selector, and profile dropdown.
`Modal` / `ConfirmDialog`	`components/common/Modal.tsx`	Standard modal, centered dialog	Provides accessible focus trapping, blurred backdrop overlays, and escape key handling.
`OperationalPagination`	`components/operations/OperationalPagination.tsx`	Paged data table controls	Standardizes pagination buttons, item counts, and page sizing for staff tables.

COMPONENT NAME	SOURCE LOCATION	PRIMARY VARIANTS	UI CONSISTENCY BENEFIT
`Toast`	`components/common/Toast.tsx`	`success`, `error`, `info`	Delivers non-disruptive feedback banners floating in the top-right corner.

3. Role-Based UI Architecture

3.1 The 5 Top-Level Application Layouts

The prototype segregates user experiences across five top-level React layouts defined in `src/app/layouts/`:

- **`PublicLayout`:** Editorial, full-width presentation container with transparent-to-solid sticky navbar and luxury brand footer. Wraps public discovery screens (Landing, Packages, Package Details).
- **`AuthLayout`:** Focused, centered card container against warm canvas background for sign-in, registration, OTP verification, and password recovery.
- **`CustomerLayout`:** Dedicated customer workspace featuring the top navigation bar, active stay breadcrumbs, customer quick-links, and profile dropdown.
- **`OperationalLayout`:** High-efficiency staff operational layout with a fixed 260px left sidebar, resort operational clock, role-specific navigation hierarchy, and responsive workspace. Used by Receptionist, Therapist, and F&B Staff.
- **`ManagerAdminLayout`:** Analytical and governance operational shell supporting KPI summary headers, date range pickers, export triggers, and delegated-capability indicators. Used by Manager and Administrator.

3.2 Actor UI Breakdown & Information Boundaries

ROLE	PRIMARY LAYOUT	PERMITTED SCREENS	STRICTLY RESTRICTED DATA	TYPICAL NAVIGATION FLOW
Customer	`CustomerLayout`	Screens 2, 7–9, 13–19, 25–29, 34–37, 41 (read-only), 45	Internal staff operations, other guests' records, resort analytics, administration.	Dashboard ➔ Retreat Booking ➔ Health Profile ➔ Spa Schedule ➔ Meals ➔ Folio ➔ Review
Receptionist	`OperationalLayout`	Screens 20–24, 30, 41–44	Detailed medical records, kitchen recipes, system feature flags, executive revenue metrics.	Dashboard ➔ Daily Arrivals ➔ Check-In & Residence ➔ Villa Assign ➔ Folio Settlement
Therapist / Instructor	`OperationalLayout`	Screens 31–33	Dietary/food allergy notes, customer payment amounts, folio charges, room billing status.	Dashboard ➔ Daily Schedule ➔ Session Detail ➔ View Precautions ➔ Complete Session
F&B Staff / Chef	`OperationalLayout`	Screens 38–40	Customer physical medical history, spa notes, passport numbers, billing records.	Dashboard ➔ Meal Prep List ➔ Order Detail ➔ Review Allergen Warnings ➔ Mark Ready
Manager	`ManagerAdminLayout`	Screens 46–48, 55, 56 (+ delegated Screens 50, 51)	Individual customer health records, raw credit card data, staff password management, security audit log (unless delegated).	Dashboard ➔ Revenue Analysis ➔ Occupancy Report ➔ Therapist Utilization ➔ System Overview
Administrator	`ManagerAdminLayout`	Screens 10–12, 49–54, 56	Clinical health details (system admin can view audit logs of access, but does not browse personal health profiles without audit trail).	Admin Dashboard ➔ Staff Management ➔ Master Catalogs ➔ Feature Flags ➔ Delegation ➔ Audit Log

> **Architectural Security Principle: Frontend Visibility != System Authorization**
>
> The front-end client utilizes `RoleGuard` to route users to contextually appropriate screens and conditionally conceal restricted controls. **However, UI-level hiding does not constitute a security boundary.** In production, every API query and mutation must enforce server-side Role-Based Access Control (RBAC), verifying user tokens, active stay tenure, and least-privilege scoping.

4. Master Screen & Navigation Overview

The official screen inventory of HSWRMS consists of exactly **56 baseline screens** catalogued in `Screen_List.md`. In the current frontend prototype, 100% of these screens are implemented as dedicated React components, registered in the router, and connected through simulated domain contexts.

4.1 Master Screen Inventory Table (Screens 1 through 56)

#	SCREEN NAME	MODULE	ACTOR	ROUTE	REACT COMPONENT	PREVIOUS SCREEN	NEXT SCREEN
1	Public Landing	Common / M1-M2	PUBLIC	/	LandingPage	External / Browser	Screen 3, 5, 13
2	Customer Dashboard	M1 – Auth	CUSTOMER	/customer/dashboard	CustomerDashboardPage	Screen 5 (Login)	Screen 8, 18, 25, 34
3	Register	M1 – Auth	CUSTOMER	/register	RegisterPage	Screen 1, 5	Screen 4
4	Email OTP Verification	M1 – Auth	CUSTOMER	/register/verify	EmailOtpVerificationPage	Screen 3	Screen 5, 8
5	Login	M1 – Auth	PUBLIC	/login	LoginPage	Screen 1, 3	Screen 2, 20, 31, 38, 49
6	Password Reset	M1 – Auth	PUBLIC	/password-reset	PasswordResetPage	Screen 5	Screen 5
7	Account Settings	M1 – Auth	CUSTOMER	/customer/account	AccountSettingsPage	Screen 2	Screen 2, 8
8	Health & Dietary Profile	M1 – Health	CUSTOMER	/customer/health-profile	HealthProfilePage	Screen 2, 4	Screen 9, 25, 34
9	Health Data Deletion Request	M1 – Health	CUSTOMER	/customer/health-profile/deletion	HealthDataDeletionPage	Screen 8	Screen 8
10	Staff Account List	M1 – Staff	ADMIN	/admin/staff	StaffAccountListPage	Screen 49	Screen 11
11	Staff Account Detail	M1 – Staff	ADMIN	/admin/staff/:id	StaffAccountDetailPage	Screen 10	Screen 10
12	Master Data Management	M1 – Master	ADMIN	/admin/master-data	MasterDataPage	Screen 49	Screen 49
13	Retreat Package List	M2 – Booking	CUSTOMER/PUBLIC	/retreats	RetreatListPage	Screen 1, 2	Screen 14
14	Retreat Package Detail	M2 – Booking	CUSTOMER/PUBLIC	/retreats/:id	RetreatDetailPage	Screen 13	Screen 15
15	Retreat Booking Form	M2 – Booking	CUSTOMER	/customer/booking/new	RetreatBookingPage	Screen 14	Screen 16
16	Booking Summary & Deposit	M2 – Booking	CUSTOMER	/customer/booking/summary	BookingSummaryPage	Screen 15	VNPay Gateway (Ext)
17	Deposit Payment Result	M2 – Booking	CUSTOMER	/customer/booking/deposit-result	DepositPaymentResultPage	VNPay Gateway (Ext)	Screen 18, 19
18	My Bookings	M2 – Booking	CUSTOMER	/customer/bookings	MyBookingsPage	Screen 2, 17	Screen 19
19	Booking Detail & Itinerary	M2 – Booking	CUSTOMER	/customer/bookings/:id	BookingDetailPage	Screen 18	Screen 25, 34, 41
20	Receptionist Dashboard	M2 – Front Desk	RECEPTIONIST	/receptionist/dashboard	ReceptionistDashboardPage	Screen 5	Screen 21, 24, 41
21	Daily Arrivals	M2 – Check-In	RECEPTIONIST	/receptionist/arrivals	DailyArrivalsPage	Screen 20	Screen 22
22	Check-In & Residence Info	M2 – Check-In	RECEPTIONIST	/receptionist/check-in/:bookingId	CheckInPage	Screen 21	Screen 23
23	Villa Assignment	M2 – Check-In	RECEPTIONIST	/receptionist/check-in/:bookingId/villa	VillaAssignmentPage	Screen 22	Screen 21, 24

#	SCREEN NAME	MODULE	ACTOR	ROUTE	REACT COMPONENT	PREVIOUS SCREEN	NEXT SCREEN
24	Villa Status Management	M2 – Accom Ops	RECEPTIONIST	/receptionist/villas	VillaStatusPage	Screen 20	Screen 20
25	Spa Treatment List	M3 – Spa	CUSTOMER	/customer/spa	SpaTreatmentListPage	Screen 2, 19	Screen 26
26	Spa Session Booking	M3 – Spa	CUSTOMER	/customer/spa/book	SpaBookingPage	Screen 25	Screen 27, 28
27	Alternative Slot Selection	M3 – Spa	CUSTOMER	/customer/spa/alternatives	SpaAlternativeSlotsPage	Screen 26 (Conflict)	Screen 28
28	Spa Booking Confirmation	M3 – Spa	CUSTOMER	/customer/spa/confirmation	SpaConfirmationPage	Screen 26, 27	Screen 29
29	My Spa Sessions	M3 – Spa	CUSTOMER	/customer/spa/sessions	MySpaSessions	Screen 2, 28	Screen 25, 26
30	Manual Extra Spa Booking	M3 – Extra Spa	RECEPTIONIST	/receptionist/spa/extra-booking	ManualExtraSpaPage	Screen 20	Screen 41
31	Therapist Dashboard	M3 – Ops	THERAPIST	/therapist/dashboard	TherapistDashboardPage	Screen 5	Screen 32
32	Therapist Daily Schedule	M3 – Ops	THERAPIST	/therapist/schedule	TherapistSchedulePage	Screen 31	Screen 33
33	Therapist Session Detail	M3 – Ops	THERAPIST	/therapist/sessions/:id	TherapistSessionDetailPage	Screen 32	Screen 32
34	Dietary Menu	M4 – Dining	CUSTOMER	/customer/dining/menu	DietaryMenuPage	Screen 2, 19	Screen 35, 36
35	Daily Meal Selection	M4 – Dining	CUSTOMER	/customer/dining/select	DailyMealSelectionPage	Screen 34	Screen 37
36	Extra A-la-carte Order	M4 – Extra F&B	CUSTOMER	/customer/dining/extra	AlaCarteOrderPage	Screen 34	Screen 37, 41
37	My Meal Plan / Orders	M4 – Dining	CUSTOMER	/customer/dining/orders	MyMealPlanPage	Screen 2, 35, 36	Screen 34
38	F&B Dashboard	M4 – Ops	F&B STAFF / CHEF	/fb/dashboard	FnbDashboardPage	Screen 5	Screen 39
39	Daily Meal Prep List	M4 – Ops	F&B STAFF / CHEF	/fb/meal-prep	MealPrepListPage	Screen 38	Screen 40
40	Meal Order Detail	M4 – Ops	F&B STAFF / CHEF	/fb/orders/:id	MealOrderDetailPage	Screen 39	Screen 39
41	Folio & Consolidated Invoice	M5 – Checkout	RECEPTIONIST / CUSTOMER	/receptionist/folio/:bookingId /customer/folio	FolioInvoicePage	Screen 20, 2	Screen 42
42	Final Payment	M5 – Checkout	RECEPTIONIST	/receptionist/checkout/:bookingId/payment /customer/checkout/payment	FinalPaymentPage	Screen 41	Screen 43
43	Final Payment Result	M5 – Checkout	RECEPTIONIST	/receptionist/checkout/:bookingId/payment-result /customer/checkout/payment-result	FinalPaymentResultPage	Screen 42	Screen 44
44	Checkout Confirmation	M5 – Checkout	RECEPTIONIST	/receptionist/checkout/:bookingId/confirmation /customer/checkout/confirmation	CheckoutConfirmationPage	Screen 43	Screen 20, 45
45	Review & Rating	M5 – Feedback	CUSTOMER	/customer/review/:bookingId	ReviewRatingPage	Screen 44 / Email	Screen 2
46	Revenue Dashboard	M5 – Analytics	MANAGER	/manager/revenue	RevenueDashboardPage	Screen 55	Screen 47, 48
47	Occupancy Report	M5 – Analytics	MANAGER	/manager/occupancy	OccupancyReportPage	Screen 55	Screen 46, 48
48	Therapist Utilization Report	M5 – Analytics	MANAGER	/manager/therapist-utilization	TherapistUtilizationReportPage	Screen 55	Screen 46, 47
49	Admin Dashboard	M6 – Admin	ADMIN	/admin/dashboard	AdminDashboardPage	Screen 5	Screen 10, 12, 50-54
50	Business Parameters	M6 – Admin	ADMIN	/admin/business-params	BusinessParametersPage	Screen 49	Screen 49

#	SCREEN NAME	MODULE	ACTOR	ROUTE	REACT COMPONENT	PREVIOUS SCREEN	NEXT SCREEN
51	Feature Flag Management	M6 – Admin	ADMIN	`/admin/feature-flags`	`FeatureFlagsPage`	Screen 49	Screen 49
52	Permission Matrix	M6 – Admin	ADMIN	`/admin/permissions`	`PermissionMatrixPage`	Screen 49	Screen 53
53	Admin Rights Delegation	M6 – Admin	ADMIN	`/admin/delegation`	`AdminDelegationPage`	Screen 49, 52	Screen 49
54	Sensitive Data Audit Log	M6 – Audit	ADMIN	`/admin/audit-log`	`SensitiveAuditLogPage`	Screen 49	Screen 49
55	Manager Dashboard	M6 – Monitoring	MANAGER	`/manager/dashboard`	`ManagerDashboardPage`	Screen 5	Screen 46-48, 56
56	System Overview Dashboard	M6 – Monitoring	ADMIN / MANAGER	`/admin/system-overview` `/manager/system-overview`	`SystemOverviewPage`	Screen 49, 55	Screen 49, 55

4.2 Shared Screens and Role-Adapted Screen Architecture

Shared vs. Role-Adapted Screen Patterns in HSWRMS

- **Public vs. Authenticated Packages (Screens 13 & 14):** When accessed by an unauthenticated visitor, `RetreatListPage` and `RetreatDetailPage` render inside `PublicLayout` with public call-to-action buttons ("Sign In to Book"). When accessed by an authenticated customer (`/customer/retreats`), they render inside `CustomerLayout` with direct booking routing.
- **Customer Folio & Consolidated Invoice (Screen 41):** In the Receptionist workspace (`/receptionist/folio/:bookingId`), Screen 41 functions as an operational checkout control deck with billing guard checks, charge reconciliation, and invoice finalization. In the Customer workspace (`/customer/folio`), Screen 41 adapts to a clean, read-only itemized statement. This achieves 100% functional coverage without inventing an unofficial screen number.
- **System Overview Dashboard (Screen 56):** Shared between Administrator and Manager. If accessed by Administrator, the top bar displays *"Administrator Clearance"* with full operational drill-down controls and police residence export affordance. If accessed by Manager, it displays *"Manager Limited Scope"* with restricted monitoring metrics.
- **External Gateway Transitions:** The prototype explicitly models simulated returns from external payment processing (VNPay deposit callback on Screen 17; VNPay final settlement callback on Screen 43) with query parameter state emulation (`?txId=...`, `?vnp_ResponseCode=00`).

5. Module 1 — Authentication & Sensitive Health Profile

5.1 Module Purpose & Scope

Module 1 governs user onboarding, identity authentication, credential recovery, confidential health profile capture, post-retreat health data deletion, staff directory administration, and resort master data management. It establishes the baseline security boundary for the resort management system.

5.2 Detailed Screen-by-Screen Breakdown (Screens 1 to 12)

Screen #1: Public Landing (Route: `/` | Layout: `PublicLayout` | Actor: Public)

Purpose: Primary public showcase communicating HeavenStay's quiet-luxury wellness ethos.
Main UI Areas: Editorial hero banner with typography, featured retreat packages carousel, holistic wellness pillars (Sleep, Nutrition, Herbal Therapies, Movement), resort sanctuary imagery, and multi-language switcher.
Primary Actions: "Khám Phá Các Gói Kỳ Nghỉ" (Explore Retreats ➔ Screen 13), "Hành Trình Chữa Lành" (Wellness Philosophy), "Đặt Kỳ Nghỉ" (Book Now ➔ Screen 13/5).
Validation & States: Language state persists via i18next across 5 languages (vi, en, zh-CN, ja, fr).
Privacy Considerations: No personal or session data exposed. Completely open to web crawlers with SEO meta tags.

Screen #2: Customer Dashboard (Route: `/customer/dashboard` | Layout: `CustomerLayout` | Actor: Customer)

Purpose: Central authenticated command center for the in-house or booked resort customer.
Main UI Areas: Personalized greeting banner ("Chào buổi sáng, Eleanor Vance"), active retreat booking summary card (Dates, Villa #102, Deposit Status, Ref HSW-88192), quick-access readiness cards for Health Profile, Spa Scheduling, and Personalized Dining.
Primary Actions: "Xem Chi Tiết & Lịch Trình" (Screen 19), "Cập Nhật Hồ Sơ Sức Khỏe" (Screen 8), "Đặt Liệu Trình Spa" (Screen 25), "Chọn Thực Đơn" (Screen 34), "Hóa Đơn Lưu Trú" (Screen 41).
Validation: RoleGuard verifies active role is 'customer'. If booking is empty, renders friendly discovery empty state.

Screen #3: Register & Screen #4: Email OTP Verification (Routes: `/register`, `/register/verify`)

Purpose: Self-service customer account creation followed by mandatory two-factor email ownership verification.
Main Controls & Inputs: Full legal name, email address, international phone number, password with strength meter, terms acceptance checkbox, 6-digit numeric OTP entry with countdown resend timer.
Validation Rules: Password requires min 8 characters, uppercase, lowercase, numeric and special characters. OTP enforces exact 6 digits with simulated 60-second cooldown.
Business Rule: Unverified accounts cannot book retreats or record health profiles. Simulated SendGrid OTP is pre-filled as `123456` for demo convenience.

Screen #5: Login & Screen #6: Password Reset (Routes: `/login`, `/password-reset`)

Purpose: Credential authentication with simulated Google/Facebook SSO buttons; self-service password recovery.
UI Controls: Email, password, "Remember me", Social SSO buttons, "Forgot Password?" hyperlink.
Validation: Validates RFC email format and non-empty password. Failed login displays localized alert banner.

Screen #8: Health & Dietary Profile (Route: `/customer/health-profile` | Actor: Customer)

Purpose: Confidential wellness profile collection covering physical conditions, mobility restrictions, dietary preferences, and food allergies.
Core Privacy Guard (Vietnam Decree on Personal Data Protection): Explicit, **unchecked-by-default consent checkbox** stating: *"Tôi đồng ý cho phép HeavenStay thu thập và xử lý thông tin sức khỏe và chế độ ăn của tôi nhằm mục đích cá nhân hóa liệu trình và đảm bảo an toàn..."*
Main UI Areas: Medical condition tags (Hypertension, Lumbar tension, Pregnancy), Mobility requirements (Ground floor, wheelchair access), Food Allergies (Peanuts, Shellfish, Gluten, Dairy), Stress level indicator (Mild, Moderate, High).
Data Minimization Boundary: Health notes are stored separately from general profile data and are never exposed to Front Desk receptionists.

Screen #9: Health Data Deletion Request (Route: `/customer/health-profile/deletion` | Actor: Customer)

Purpose: Customer affordance to exercise the "Right to be Forgotten" regarding sensitive health and dietary history post-retreat.
UI Areas: Deletion eligibility notice (requires completed retreat checkout), confirmation input requiring customer to type "CONFIRM", impact summary warning that future retreats will require re-entry.
Business Rule: Health data deletion soft-deletes or anonymizes medical tags while retaining statutory financial records (M5 Folio) for accounting compliance.

Screens #10, #11, #12: Staff Directory & Master Data Management (Routes: `/admin/staff`, `/admin/master-data`)

Purpose: Administrator maintenance of resort staff accounts, role assignments (Receptionist, Therapist, Chef, Manager), and resort master data catalogs (Villa Categories, Spa Services, Retreat Packages).
Controls: Searchable staff data table with status pills (Active, Suspended), role dropdown, master catalog tabs with inline editing of package durations and service pricing.

6. Module 2 — Retreat Booking & Accommodation

6.1 Customer Booking Flow (Screens 13 to 19)

Module 2 governs the primary revenue lifecycle of HSWRMS, progressing from catalog exploration to front-desk check-in:

> **Customer Journey:** Screen 13 (Retreat Package Catalog) ➔ Screen 14 (Package Detail & Included Services) ➔ Screen 15 (Booking Form: Dates, Guests, Villa Type) ➔ Screen 16 (Booking Summary & Deposit Calculation) ➔ External Gateway (VNPay Deposit 30%) ➔ Screen 17 (Deposit Payment Result) ➔ Screen 18 (My Bookings) ➔ Screen 19 (Booking Detail & Comprehensive Itinerary).

6.2 Key Domain Concept: Villa Category vs. Physical Villa

> **Critical Hospitality Principle: Catalog Category != Physical Inventory Unit**
>
> In HSWRMS, the Customer books a **Villa Category** (e.g. *Ocean Pool Villa*, *Tranquil Garden Villa*, *Mountain Sunrise Suite*) during the online reservation flow (Screen 15). The Customer **never** reserves a physical villa number directly online.
>
> The assignment of a **Physical Villa** (e.g. *Villa #102*, *Villa #205*) occurs exclusively at the Front Desk during the Check-In process (Screen 23) managed by the **Receptionist**. This guarantees that housekeeping status, maintenance blocks, and early arrivals can be dynamically managed without double-booking rooms.

6.3 Receptionist Front Desk Operations (Screens 20 to 24)

- **Screen 20 (Receptionist Dashboard):** Operational morning briefing showing daily arrivals count, expected departures, in-house guests, and real-time villa occupancy.
- **Screen 21 (Daily Arrivals):** Searchable arrivals queue displaying guest names, arrival times, deposit status, and pre-assigned villa categories. Includes the statutory compliance banner for daily residence declarations.
- **Screen 22 (Check-In & Residence Information):** Captures mandatory identification data pursuant to the **Vietnam Law on Residence 2020** (Luật Cư trú 2020 & Nghị định 144/2021/NĐ-CP). Collects full legal name, nationality, passport/CCCD number, issue date, permanent address, and entry transport.
- **Screen 23 (Villa Assignment):** Displays real-time physical villa grid filtered by matching category. Receptionist verifies housekeeping readiness (Available vs Cleaning vs Maintenance) and binds a physical villa to the active booking.
- **Screen 24 (Villa Status Management):** Housekeeping operational matrix enabling Receptionist and cleaning staff to transition villas between *Available*, *Occupied*, *Vacant Needs Cleaning*, and *Maintenance*.

7. Module 3 — Spa & Therapy Scheduling

7.1 Approved Conceptual Scheduling Logic: System-Driven Candidate Generation

Authoritative M3 Scheduling Architecture (Approved Business Flow)

Under the latest approved business logic for HSWRMS:

1. The Customer (or Receptionist) specifies only: **Treatment Desired**, **Preferred Date / Time Window**, and **Optional Notes**.
2. **HSWRMS generates the complete scheduling candidate:**
   `Time Slot + Qualified Therapist / Instructor + Compatible Treatment Room`.
   The user does **not** manually select or assemble these operational resources individually.
3. **Atomic 2D Reservation:** Both the qualified therapist and the treatment room are locked atomically in a single database transaction. No partial reservation is ever permitted.
4. **Revalidation at Confirmation:** Before finalizing confirmation, HSWRMS revalidates resource availability to eliminate race conditions.
5. **Recovery & Replacement:** If a therapist becomes unavailable (e.g. sick leave), HSWRMS attempts an invisible same-slot qualified therapist substitution. If no practitioner is available, the customer is invited to consent to a rescheduled time or alternate treatment.

7.2 Detailed Screen-by-Screen Breakdown (Screens 25 to 33)

#	SCREEN NAME	ACTOR	PRIMARY OPERATIONAL PURPOSE	KEY UI CONTROLS & DATA DISPLAYED
25	Spa Treatment List	Customer	Browse therapeutic catalog; view included retreat treatments vs. extra chargeable therapies.	Category filter pills (Massage, Herbal, Hydrotherapy, Yoga), duration badges, included indicators ("Trong Gói"), pricing, "Đặt Liệu Trình" CTA.
26	Spa Session Booking	Customer	Input preferred date, time window, and initiate atomic candidate matching.	Date strip (7-day horizon), time slot grid, special request textarea, candidate preview tile showing assigned room suite and practitioner.
27	Alternative Slot Selection	Customer	Present system-recommended alternatives when requested slot has resource conflicts.	Conflict diagnosis banner (Therapist busy / Room occupied), curated alternative recommendation cards with one-click selection.
28	Spa Booking Confirmation	Customer	Display confirmed 2D lock reservation, appointment time, room location, and calendar sync.	Success checkmark, session voucher summary, simulated Google Calendar sync toggle, reminder schedule notice.
29	My Spa Sessions	Customer	Review upcoming and past therapy sessions; view confirmation countdown status.	Timeline cards sorted chronologically, session status chips (Scheduled, Confirmed, Completed), room name, cancellation action.
30	Manual Extra Spa Booking	Receptionist	Book an ad-hoc extra spa service for an in-house customer with automatic charge posting to folio.	Customer lookup dropdown, treatment picker, system candidate slot generator, automatic charge posting preview (`M3 ➔ M5 Folio`).
31	Therapist Dashboard	Therapist	Personal daily overview showing assigned treatments, today's schedule, and room assignments.	Next session highlight card, daily KPI metrics (Sessions today, Total therapy minutes), quick link to full daily schedule.
32	Therapist Daily Schedule	Therapist	Chronological agenda of treatments for selected date; access individual session details.	Timeline schedule view, customer first name, treatment type, room suite, status chips, "Chi Tiết" button.
33	Therapist Session Detail	Therapist	Review treatment safety precautions; record clinical notes; mark session Completed or No-Show.	**Treatment Safety Health Notes** (muscular/joint precautions only), special requests, clinical notes input, action buttons (Bắt Đầu, Hoàn Tất, Vắng Mặt).

7.3 Dual State Model: Session Status vs. Confirmation Status

A critical architectural requirement of M3 is the separation of the **Operational Session Lifecycle** from the **Customer Confirmation Protocol**:

Session Operational Status (Entity Lifecycle)

- `SCHEDULED`: Initial system-generated reservation locked in database.
- `CUSTOMER_CONFIRMED`: Customer responded positively to pre-session confirmation.
- `RESCHEDULING`: In process of finding alternative slot due to disruption.
- `CANCELLED`: Session revoked by customer or resort administration.
- `COMPLETED`: Treatment successfully delivered by practitioner.
- `NO_SHOW`: Customer failed to attend session within grace period.

Confirmation Status (Customer Engagement Protocol)

- `PENDING`: Confirmation request dispatched; awaiting customer response.
- `CONFIRMED`: Customer clicked confirm in app or notification link.
- `DECLINED`: Customer actively requested cancellation or reschedule.
- `EXPIRED`: Confirmation window lapsed without customer response.
  Critical Rule: EXPIRED != NO_SHOW. An expired confirmation allows the resort to reallocate resources or follow up, whereas NO_SHOW occurs only after session start time.

7.4 Reminder Timeline Specification

TRIGGER TIMELINE	MESSAGE TYPE	PRIMARY CHANNEL	SYSTEM ACTION & BEHAVIORAL IMPACT
T - 24 Hours	Informational Reminder	In-App & Email	Notifies customer of upcoming treatment tomorrow; displays preparation guidelines (e.g. hydration, attire).
T - 2 Hours	Confirmation Request	In-App Banner & SMS/Push	Requests active confirmation from customer (Confirm / Reschedule). Confirmation status transitions to `PENDING`.

TRIGGER TIMELINE	MESSAGE TYPE	PRIMARY CHANNEL	SYSTEM ACTION & BEHAVIORAL IMPACT
T - 75 Minutes	Final Reminder	High-priority Alert	Sent if confirmation remains pending with 15 minutes remaining before release deadline.
T - 60 Minutes	Confirmation Deadline	System Event	If unconfirmed, status transitions to `EXPIRED`. System may alert Receptionist to call villa or reassign therapist.
T - 15 Minutes	Upcoming Reminder	In-App Notification	Advises customer to proceed to the Spa Sanctuary treatment pavilion.

7.5 Sensitive Health Data Isolation: Treatment Safety Health Profile

Therapist Data Minimization Boundary (Screen 33)

Therapists have **no access** to the customer's full medical history, bloodwork, medications, psychological assessments, home address, passport data, or food allergies. Screen 33 renders a strictly filtered `TherapistSafeWellnessView` projecting only:

- Customer first name only (e.g. "Eleanor").
- Musculoskeletal conditions (e.g. "Mild lumbar tension", "Past knee surgery").
- Cardiovascular precautions affecting heat/pressure therapies (e.g. "Mild hypertension (well managed)").
- Customer physical comfort preferences (e.g. "Prefers lavender oil; lower back cushion").

8. Module 4 — Dietary F&B Management

8.1 Customer Dietary Journey (Screens 34 to 37)

Module 4 delivers bespoke organic dining tailored to each customer's nutritional goals and medical food intolerances:

- **Screen 34 (Dietary Menu):** Interactive dining catalog showcasing farm-to-table dishes categorized by meal period (Breakfast, Lunch, Dinner, All-Day) and wellness tags (Pescatarian, Vegan, Gluten-Conscious, Ayurvedic).
- **Screen 35 (Daily Meal Selection):** Allows customer to pre-select included package meals for each stay date, enabling kitchen staff to plan fresh ingredient prep ahead of time.
- **Screen 36 (Extra A-la-carte Order):** Enables ordering supplemental gourmet wellness dishes, cold-pressed elixirs, or artisanal tea service charged directly to the Customer Folio (`M4 ➔ M5 Folio`).
- **Screen 37 (My Meal Plan / Orders):** Aggregated view displaying selected daily courses, fulfillment status (Pending, Preparing, Ready, Delivered), and submitted extra charges.

8.2 Kitchen & Chef Operational Screens (Screens 38 to 40)

- **Screen 38 (F&B Dashboard):** Kitchen command board summarizing upcoming meal service periods, total daily covers, critical allergy alerts, and active kitchen ticket counts.
- **Screen 39 (Daily Meal Prep List):** Workload aggregation board grouping meal prep tickets by meal period, villa delivery destination, and dietary requirements. High-contrast warning chips highlight severe allergies (Nuts, Shellfish, Eggs).
- **Screen 40 (Meal Order Detail):** Individual preparation ticket displaying dish recipes, plating instructions, customer special requests, and status transition triggers (Preparing ➔ Ready ➔ Delivered).

8.3 Personalized Menu Formulation Logic

> **Dynamic Menu Availability Equation**
>
> ```
> [Customer Allergy/Dietary Profile] + [Dish Ingredient Formulation] + [Real-Time Stock Availability]
>       = Personalized Available Menu (Unsafe items masked or flagged with warnings)
> ```

8.4 Advanced Ingredient Quantity Model & Inventory Recalculation

In the latest approved requirements, culinary operations rely on a four-tier relational model linking menu items to physical storeroom inventory:

Relational Structure: `MENU_ITEM` ──(1:N)──► `MENU_ITEM_INGREDIENT` ──(N:1)──► `INGREDIENT` ──(1:N)──► `INGREDIENT_BATCH`
Quantity Computation: `quantity_per_serving × number_of_servings = required_ingredient_quantity`

Estimated Ingredient Requirement: Calculated in advance when customers submit Daily Meal Selections (Screen 35), allowing the Head Chef to forecast daily requisition orders.

Actual Ingredient Usage: Recorded upon fulfillment on Screen 40, accounting for yield loss, culinary adjustments, and batch expiration FIFO tracking.

> **Architecture Synchronization Note: Current Prototype vs. Relational Model**
>
> In the current React prototype (`types/dining.ts`), dietary options are implemented via static tag arrays (`dietaryTags`, `allergenTags`) and mock orders. The detailed relational entity `MENU_ITEM_INGREDIENT` and batch inventory math represent an approved requirement scheduled for production backend implementation and is flagged in the Gap Analysis.

9. Module 5 — Consolidated Checkout & Analytics

9.1 The Checkout Journey (Screens 41 to 45)

Module 5 provides end-of-stay financial reconciliation and management performance analytics:

> **Checkout Sequence:** Screen 41 (Customer Folio & Consolidated Invoice) ➔ Screen 42 (Final Payment Console) ➔ Screen 43 (Final Payment Result Callback) ➔ Screen 44 (Checkout Confirmation & Departure Handover) ➔ Screen 45 (Review & Rating).

9.2 The Critical Billing Guard & Checkout Blocker

> **Mandatory Business Rule: Outstanding Services Block Resort Checkout (Screen 44)**
>
> Checkout **cannot** be completed while unpaid or unfinalized charges remain on the customer folio. The billing guard evaluates four distinct blocker conditions:
>
> 1. `unpaid_balance`: Remaining retreat package balance or unbilled extra charges greater than 0 VND.
> 2. `unsettled_spa`: Extra spa treatments delivered but not yet posted or settled on the folio.
> 3. `unsettled_fnb`: In-villa dining or a-la-carte orders in progress or unbilled.
> 4. `not_in_stay`: Reservation not in active checked-in stay status.
>
> If any blocker is triggered, Screen 44 displays the **Resort Departure Handover Blocked** panel with itemized amounts, preventing checkout until resolved.

9.3 Fundamental Domain Rule: Payment Success != Checkout Complete

A recurring trap in generic hotel software is conflating card payment authorization with resort checkout. In HSWRMS:

PAYMENT SUCCESS (SCREEN 43)	CHECKOUT COMPLETED (SCREEN 44)
A purely financial transaction outcome returned by the VNPay payment gateway or cash receipt confirmation.	An operational state transition across multiple resort entities.
Reduces `outstandingBalance` to 0 VND.	Requires 4 coordinated actions: 1. Consolidated Invoice marked `PAID`. 2. Customer Folio marked `CLOSED`. 3. Room Booking status transitions to `CHECKED_OUT`. 4. Physical Villa status transitions to `VACANT_NEEDS_CLEANING`.

9.4 Manager Analytical Reports (Screens 46 to 48)

- **Screen 46 (Revenue Dashboard):** Aggregated gross revenue charts (Recharts) broken down by Retreat Packages, Spa Holistic Therapies, and Dining Extras. Excludes personal customer medical or payment card identifiers.
- **Screen 47 (Occupancy Report):** Accommodation analytics tracking villa category occupancy percentages, average length of stay (ALOS), and turn-around cleaning intervals.
- **Screen 48 (Therapist Utilization Report):** Workforce efficiency analytics displaying therapist session hours, billable treatment utilization percentages, and treatment category popularity.

10. Module 6 — System Administration & Governance

10.1 Module Purpose & Scope (Screens 49 to 56)

Module 6 enforces governance, operational parameters, security auditing, and least-privilege role boundaries across the entire HSWRMS ecosystem:

- **Screen 49 (Admin Dashboard):** Administrator command portal displaying system operational health, active staff counts, audit summary, and quick links to configuration consoles.
- **Screen 50 (Business Parameters):** Configurable business policy settings: Deposit percentage (30%), Retreat cancellation refund lead times (14 days, 7 days), Spa cancellation windows, and tax rates.
- **Screen 51 (Feature Flag Management):** Runtime operational toggles to enable or disable modules or sub-features (e.g. `enable_vnpay_live`, `enable_in_villa_delivery`, `enable_guest_rating_portal`).
- **Screen 52 (Permission Matrix):** Granular matrix mapping system roles to operational screens and capabilities.
- **Screen 53 (Limited Admin Rights Delegation):** Formal mechanism allowing Admin to temporarily delegate tightly scoped operational capabilities to Manager without granting full Admin credentials.
- **Screen 54 (Sensitive Data Audit Log):** Immutable security access log recording every view, update, or deletion of sensitive health profiles, identity documents, and payment records.
- **Screen 55 (Manager Dashboard):** Executive management portal providing overview of revenue, operations, and resort metrics within granted permissions.
- **Screen 56 (System Overview Dashboard):** Cross-module application health board and official statutory export interface for the **Police Residency Declaration Report** (Luật Cư trú 2020).

10.2 Role Demarcation: Administrator vs. Manager

> **Core Security Boundary: The Manager Is NOT a Full Administrator**
>
> Under HSWRMS security architecture:
>
> - **Administrator:** Technical authority responsible for account provisioning, RBAC matrix reconfiguration, security log auditing, master data schemas, and disaster recovery. Has no clinical care duties.
> - **Manager:** Operational resort supervisor possessing read-only analytical visibility over revenues, occupancy, and labor utilization. Manager **cannot** reset staff passwords, alter role matrices, access raw medical profiles, or inspect security audit logs unless explicitly granted via Screen 53.

10.3 Limited Admin Rights Delegation Mechanism (Screen 53)

When operational emergencies occur (e.g. coastal weather advisories requiring temporary adjustment of retreat cancellation windows), Administrator can grant a **Scoped Delegation**:

- Delegation must specify: Delegatee (Manager), Granted Capabilities (e.g. `configure_business_parameters`), Validity Window (e.g. 24 hours), and Mandatory Justification Reason.
- **Anti-Escalation Protection (BR-06):** Master password changes, security audit log viewing, and re-delegation powers are strictly non-delegable.
- Delegations automatically lapse at expiration and are logged in the audit trail.

11. Screen-by-Screen UI Catalog

This catalog provides a compact technical specification for all 56 official baseline screens implemented in the HSWRMS prototype. Every entry lists the verified route, component name, actor, primary controls, displayed data, primary call-to-action (CTA), and simulated status/error states.

11.1 Screens 1 to 12 (Common, Public & Module 1)

#	SCREEN & ROUTE	ACTOR	PURPOSE & MAIN CONTROLS	DISPLAYED DATA	PRIMARY CTA	STATUS / ERRORS
1	**Public Landing** `/`	PUBLIC	Editorial discovery; package cards, language switcher, brand story.	Hero statement, 3 retreat packages, wellness pillars, testimonials.	"Khám Phá Các Gói Kỳ Nghỉ"	Language toggle (vi, en, zh-CN, ja, fr)
2	**Customer Dashboard** `/customer/dashboard`	CUSTOMER	Main customer hub; active stay banner, action shortcuts.	Customer name, active booking summary, villa #102, readiness chips.	"Xem Chi Tiết & Lịch Trình"	Stay status (Confirmed, In-Stay)
3	**Register** `/register`	CUSTOMER	Account signup; input name, email, phone, password, terms.	Registration form fields, password strength indicator.	"Tạo Tài Khoản"	Validation error, email conflict
4	**Email OTP Verify** `/register/verify`	CUSTOMER	Two-factor email ownership verification; 6-digit OTP input.	Masked email address, resend cooldown timer.	"Xác Thực Tài Khoản"	Invalid OTP, expired OTP
5	**Login** `/login`	PUBLIC	Credential sign-in; email, password, Google/Facebook SSO.	Login form fields, "Remember Me" checkbox, reset link.	"Đăng Nhập"	Invalid credentials, locked account
6	**Password Reset** `/password-reset`	PUBLIC	Self-service recovery; enter email, receive simulated token.	Email field, reset instructions, new password inputs.	"Gửi Liên Kết Đặt Lại"	Email not found, weak password
7	**Account Settings** `/customer/account`	CUSTOMER	Personal details, contact phone, password change.	Name, email, phone, notification preferences.	"Lưu Thay Đổi"	Password mismatch, update success
8	**Health & Dietary Profile** `/customer/health-profile`	CUSTOMER	Confidential medical & food profile with explicit consent.	Medical conditions, mobility, allergies, dietary goals, consent status.	"Lưu Hồ Sơ Sức Khỏe"	Consent required error, save success
9	**Health Data Deletion** `/customer/health-profile/deletion`	CUSTOMER	GDPR/PDPD right to be forgotten request for health data.	Eligibility criteria, confirmation phrase field, impact warning.	"Xác Nhận Yêu Cầu Xóa"	Stay active blocker, request lodged
10	**Staff Account List** `/admin/staff`	ADMIN	Staff directory; search, filter by role/department, add staff.	Staff table: ID, name, email, role, status pill, last login.	"+ Thêm Nhân Viên"	Filter empty, pagination
11	**Staff Account Detail** `/admin/staff/:id`	ADMIN	Staff profile, role assignment, active/suspend toggle.	User credentials, department, assigned permissions, audit trail.	"Cập Nhật Tài Khoản"	Role conflict, status changed
12	**Master Data Management** `/admin/master-data`	ADMIN	Resort master catalogs; villa types, treatments, packages.	Tabbed master data tables, pricing, duration, inclusion rules.	"Lưu Cấu Hình Dữ Liệu"	Pricing validation, catalog updated

11.2 Screens 13 to 24 (Module 2: Retreat Booking & Accommodation)

#	SCREEN & ROUTE	ACTOR	PURPOSE & MAIN CONTROLS	DISPLAYED DATA	PRIMARY CTA	STATUS / ERRORS
13	**Retreat Package List** `/retreats`	CUSTOMER	Browse wellness packages; wellness goal filters.	Package cards, days/nights, included therapies, starting rate.	"Xem Chi Tiết"	Filter results empty
14	**Retreat Package Detail** `/retreats/:id`	CUSTOMER	In-depth retreat itinerary, included meals/spa, villa options.	Editorial narrative, daily schedule breakdown, inclusions list.	"Đặt Gói Nghỉ Dưỡng"	Package not found (404)
15	**Retreat Booking Form** `/customer/booking/new`	CUSTOMER	Date range selector, guest count, villa category selection.	Calendar picker, price breakdown, villa amenities.	"Tiếp Tục Đến Tóm Tắt"	Invalid dates, unavailable villa
16	**Booking Summary & Deposit** `/customer/booking/summary`	CUSTOMER	Review reservation, calculate mandatory 30% VNPay deposit.	Total package cost, 30% deposit due, cancellation terms.	"Thanh Toán Đặt Cọc (VNPay)"	Price revalidation conflict
17	**Deposit Payment Result** `/customer/booking/deposit-result`	CUSTOMER	VNPay payment callback display; confirm booking reference.	Transaction ID, amount paid, booking reference, receipt.	"Xem Lịch Trình Kỳ Nghỉ"	Payment Success vs. Failure
18	**My Bookings** `/customer/bookings`	CUSTOMER	Customer booking history; upcoming, active, completed stays.	Booking cards, stay dates, villa category, status chip.	"Chi Tiết Đặt Chỗ"	No active bookings
19	**Booking Detail & Itinerary** `/customer/bookings/:id`	CUSTOMER	Day-by-day retreat timeline; check-in details, inclusions.	Timeline cards, scheduled spa sessions, assigned meals.	"Đặt Dịch Vụ Tiếp Theo"	Booking status badge

#	SCREEN & ROUTE	ACTOR	PURPOSE & MAIN CONTROLS	DISPLAYED DATA	PRIMARY CTA	STATUS / ERRORS
20	**Receptionist Dashboard** `/receptionist/dashboard`	RECEPTIONIST	Front desk operational KPIs; arrivals, departures, villa counts.	Today's metrics, quick search, recent arrival cards.	"Xem Khách Đến Hôm Nay"	Real-time counter updates
21	**Daily Arrivals** `/receptionist/arrivals`	RECEPTIONIST	Searchable arrivals queue; statutory police declaration banner.	Arrival table: ETA, guest, booking ref, deposit paid, villa status.	"Bắt Đầu Check-In"	Expected, Arrived, Checked-In
22	**Check-In & Residence Info** `/receptionist/check-in/:id`	RECEPTIONIST	Verify guest identity; collect Vietnam Residence Law 2020 data.	Full legal name, passport/CCCD, nationality, address, entry point.	"Tiếp Tục Phân Bổ Villa"	Missing mandatory residence data
23	**Villa Assignment** `/receptionist/check-in/:id/villa`	RECEPTIONIST	Assign physical available villa matching booked category.	Physical villa cards, floor, view, readiness status.	"Xác Nhận Phân Bổ Villa"	Villa in maintenance / cleaning
24	**Villa Status Management** `/receptionist/villas`	RECEPTIONIST	Resort-wide villa grid; housekeeping status management.	Villa cards grouped by category, status toggle chips.	"Cập Nhật Trạng Thái"	Available, Occupied, Cleaning, Maint

11.3 Screens 25 to 33 (Module 3: Spa & Therapy Scheduling)

#	SCREEN & ROUTE	ACTOR	PURPOSE & MAIN CONTROLS	DISPLAYED DATA	PRIMARY CTA	STATUS / ERRORS
25	**Spa Treatment List** `/customer/spa`	CUSTOMER	Browse therapy catalog; category filter, duration, included tags.	Treatment cards, benefits, contraindications preview.	"Đặt Lịch Liệu Trình"	Filter empty
26	**Spa Session Booking** `/customer/spa/book`	CUSTOMER	Select date, time slot; system computes 2D resource candidate.	7-day date strip, time slot grid, assigned room & practitioner tile.	"Xác Nhận Giữ Chỗ"	Resource conflict (➔ Screen 27)
27	**Alternative Slot Selection** `/customer/spa/alternatives`	CUSTOMER	Select alternative available slots after scheduling conflict.	Conflict diagnosis banner, suggested alternative slot cards.	"Chọn Khung Giờ Này"	All alternatives booked error
28	**Spa Booking Confirmation** `/customer/spa/confirmation`	CUSTOMER	Confirmed appointment summary; calendar sync, reminder notices.	Confirmed voucher, room pavilion, practitioner, reminder times.	"Xem Buổi Trị Liệu Của Tôi"	Atomic lock confirmed
29	**My Spa Sessions** `/customer/spa/sessions`	CUSTOMER	Chronological list of booked treatments; cancel/reschedule.	Session cards, confirmation status chip, room, cancel modal.	"Chi Tiết / Hủy Lịch"	Scheduled, Confirmed, Completed
30	**Manual Extra Spa Booking** `/receptionist/spa/extra-booking`	RECEPTIONIST	Front desk books extra spa for customer; posts charge to folio.	Customer picker, treatment selector, folio posting preview.	"Xác Nhận & Ghi Vào Hóa Đơn"	Candidate unavailable, charge posted
31	**Therapist Dashboard** `/therapist/dashboard`	THERAPIST	Practitioner overview; next session highlight, daily session count.	Next client card, room suite, daily treatment hours metric.	"Xem Lịch Làm Việc Hôm Nay"	Duty status (Available, Busy)
32	**Therapist Daily Schedule** `/therapist/schedule`	THERAPIST	Timeline agenda for date; session cards with customer first names.	Timeline blocks, treatment types, assigned rooms, session chips.	"Xem Chi Tiết & Ghi Chú"	Scheduled, Confirmed, Completed
33	**Therapist Session Detail** `/therapist/sessions/:id`	THERAPIST	Review **Treatment Safety Notes**; mark Completed/No-Show.	Muscular/joint precautions, clinical notes input, action buttons.	"Hoàn Tất Buổi Trị Liệu"	Completed vs. No-Show

11.4 Screens 34 to 40 (Module 4: Dietary F&B Management)

#	SCREEN & ROUTE	ACTOR	PURPOSE & MAIN CONTROLS	DISPLAYED DATA	PRIMARY CTA	STATUS / ERRORS
34	**Dietary Menu** `/customer/dining/menu`	CUSTOMER	Browse organic menu; filter by meal period & wellness tags.	Dish cards, nutrition info, dietary tags, allergen badges.	"Chọn Cho Hôm Nay"	Allergen conflict masked
35	**Daily Meal Selection** `/customer/dining/select`	CUSTOMER	Pre-select included package meals for breakfast, lunch, dinner.	Meal period tabs, course selectors (starter, main, dessert).	"Xác Nhận Thực Đơn Ngày"	Pre-selection confirmed
36	**Extra A-la-carte Order** `/customer/dining/extra`	CUSTOMER	Order supplemental gourmet dishes charged to customer folio.	Priced dish catalog, cart drawer, villa delivery time field.	"Đặt Món & Ghi Vào Hóa Đơn"	Cart empty, charge posted to M5
37	**My Meal Plan / Orders** `/customer/dining/orders`	CUSTOMER	Review selected daily courses and submitted a-la-carte orders.	Ordered dishes, fulfillment status pill, itemized extra amounts.	"Xem Thực Đơn Thêm"	Pending, Preparing, Ready, Delivered
38	**F&B Dashboard** `/fb/dashboard`	F&B STAFF / CHEF	Kitchen command board; upcoming covers, active orders, allergy alerts.	Service period cards, active ticket count, severe allergen alerts.	"Xem Danh Sách Chế Biến"	Real-time kitchen load

#	SCREEN & ROUTE	ACTOR	PURPOSE & MAIN CONTROLS	DISPLAYED DATA	PRIMARY CTA	STATUS / ERRORS
39	**Daily Meal Prep List** `/fb/meal-prep`	F&B STAFF / CHEF	Aggregated culinary prep workload grouped by meal period & villa.	Dish prep cards, portion counts, prominent allergen warning chips.	"Xem Chi Tiết Món Ăn"	Filter by service period
40	**Meal Order Detail** `/fb/orders/:id`	F&B STAFF / CHEF	Individual culinary ticket; recipe notes, transition to Ready.	Customer name, villa #, ingredient notes, status toggle buttons.	"Đánh Dấu: Đã Sẵn Sàng"	Preparing ➔ Ready ➔ Delivered

11.5 Screens 41 to 48 (Module 5: Checkout & Analytics)

#	SCREEN & ROUTE	ACTOR	PURPOSE & MAIN CONTROLS	DISPLAYED DATA	PRIMARY CTA	STATUS / ERRORS
41	**Folio & Consolidated Invoice** `/receptionist/folio/:id` `/customer/folio`	RECEPTIONIST CUSTOMER	Itemized statement of package, spa extras, and dining extras.	Package balance, spa line items, dining line items, outstanding due.	"Tiến Hành Thanh Toán" (Staff)	Pending Settlement, Settled
42	**Final Payment** `/receptionist/checkout/:id/payment`	RECEPTIONIST	Final settlement payment console; VNPay gateway simulation.	Total settlement due, booking ref, VNPay QR / mode selector.	"Xác Nhận Thanh Toán (VNPay)"	Processing, Gateway timeout
43	**Final Payment Result** `/receptionist/checkout/:id/payment-result`	RECEPTIONIST	Transaction outcome callback; balance marked zero.	Transaction ID, amount paid, status checkmark, print button.	"Tiến Hành Trả Phòng (Check-Out)"	Payment Success vs. Failure
44	**Checkout Confirmation** `/receptionist/checkout/:id/confirmation`	RECEPTIONIST	Finalize departure; billing guard checks; release villa to cleaning.	Billing guard verification, checkout record, housekeeping update.	"Hoàn Tất Trả Phòng & Bàn Giao"	Blocked by Unpaid Charges
45	**Review & Rating** `/customer/review/:bookingId`	CUSTOMER	Post-checkout retreat feedback; star rating, textual comments.	5-star rating selector, category ratings (Spa, Food, Villa), text area.	"Gửi Đánh Giá Kỳ Nghỉ"	Optional; submitted confirmation
46	**Revenue Dashboard** `/manager/revenue`	MANAGER	Gross revenue trends; breakdown by Package, Spa, and Dining.	Bar charts, donut distribution, gross revenue totals, date filter.	"Xuất Báo Cáo Doanh Thu"	Date range filter update
47	**Occupancy Report** `/manager/occupancy`	MANAGER	Accommodation analytics; occupancy rates by villa category.	Occupancy percentage cards, trend graphs, average length of stay.	"Xuất Báo Cáo Công Suất"	Historical filter
48	**Therapist Utilization** `/manager/therapist-utilization`	MANAGER	Workforce efficiency; billable session hours per therapist.	Practitioner utilization bars, total therapy hours delivered.	"Xuất Báo Cáo Trị Liệu"	Utilization benchmark thresholds

11.6 Screens 49 to 56 (Module 6: Administration & Governance)

#	SCREEN & ROUTE	ACTOR	PURPOSE & MAIN CONTROLS	DISPLAYED DATA	PRIMARY CTA	STATUS / ERRORS
49	**Admin Dashboard** `/admin/dashboard`	ADMIN	System administration command center; security summary.	Active users, audit event counts, flag status, configuration links.	"Quản Trị Người Dùng & Cấu Hình"	System health status
50	**Business Parameters** `/admin/business-params`	ADMIN	System business settings; deposit rates, cancellation windows.	Deposit % (30%), refund cutoff days, spa cancellation lead hours.	"Lưu Cấu Hình Tham Số"	Validation error, saved success
51	**Feature Flag Management** `/admin/feature-flags`	ADMIN	Runtime toggles for system features; emergency circuit breakers.	Feature list: VNPay live, In-villa dining, Rating portal, Audit log.	"Cập Nhật Cờ Tính Năng"	Protected flag warning
52	**Permission Matrix** `/admin/permissions`	ADMIN	Role-to-screen & role-to-capability RBAC access table.	Matrix grid mapping 6 roles to operational modules and endpoints.	"Lưu Ma Trận Phân Quyền"	Admin role uneditable
53	**Admin Rights Delegation** `/admin/delegation`	ADMIN	Delegate limited administrative capabilities to Manager.	Active delegations table, capability picker, justification field.	"+ Cấp Quyền Ủy Quyền Mới"	Anti-escalation guard, Expired
54	**Sensitive Audit Log** `/admin/audit-log`	ADMIN	Immutable audit log of all sensitive health & financial access.	Timestamp, actor ID, action type (View, Update, Export), target record.	"Xuất Nhật Ký Kiểm Toán"	Search / filter results
55	**Manager Dashboard** `/manager/dashboard`	MANAGER	Executive operational hub for resort operations manager.	Revenue KPIs, today's occupancy, active therapies, culinary count.	"Xem Chi Tiết Báo Cáo"	Read-only manager scope
56	**System Overview Dashboard** `/admin/system-overview`	ADMIN MANAGER	Cross-module application health & statutory police residence export.	Accommodation, therapy, kitchen, and folio operational health.	"Xuất Báo Cáo Khai Báo Cư Trú"	Admin clearance vs. Manager scope

12. Status & State Reference Guide

12.1 Persistent Database States vs. Transient Runtime Results

> **Architectural Distinction: Persistent State vs. Runtime Event**
> To maintain data integrity, developers and evaluators must distinguish **Persistent States** (stored in database entities and surviving server restarts) from **Transient Runtime Results** (ephemeral evaluation outcomes returned during request processing, such as an invalid payment checksum or a scheduling resource collision). Runtime events must never be persisted as entity statuses.

12.2 Consolidated System Status Matrix

ENTITY / DOMAIN	STATUS VALUE	TYPE	BUSINESS MEANING & TRANSITION TRIGGER	UI REPRESENTATION
Retreat Booking (`Room_Booking`)	`PENDING_DEPOSIT`	Persistent	Reservation initiated; awaiting VNPay 30% deposit within 15 min.	CHỜ ĐẶT CỌC
	`CONFIRMED`	Persistent	Deposit received; sanctuary villa category guaranteed.	ĐÃ XÁC NHẬN
	`CHECKED_IN`	Persistent	Customer verified at Front Desk; physical villa assigned (Screen 22).	ĐANG LƯU TRÚ
	`CHECKED_OUT`	Persistent	Folio settled; departure handover complete (Screen 44).	ĐÃ TRẢ PHÒNG
Physical Villa (`Villa_Inventory`)	`AVAILABLE`	Persistent	Clean, inspected, ready for immediate guest check-in.	SẴN SÀNG (Green dot)
	`OCCUPIED`	Persistent	In-house customer residing in villa.	ĐANG Ở (Amber dot)
	`VACANT_CLEANING`	Persistent	Customer departed; assigned to housekeeping for sanitization.	CẦN DỌN DẸP (Purple dot)
	`MAINTENANCE`	Persistent	Out of order for mechanical, pool, or structural repair.	BẢO TRÌ (Red dot)
Spa Session (`Spa_Session`)	`SCHEDULED`	Persistent	Candidate generated; therapist and room atomically locked.	ĐÃ LÊN LỊCH
	`CUSTOMER_CONFIRMED`	Persistent	Customer affirmed upcoming attendance via pre-session prompt.	KHÁCH XÁC NHẬN
	`RESCHEDULING`	Persistent	Disruption detected; alternative slot negotiation in progress.	ĐANG ĐỔI LỊCH
	`CANCELLED`	Persistent	Session revoked prior to treatment start time.	ĐÃ HỦY
	`COMPLETED`	Persistent	Therapist marked session delivered; clinical notes recorded.	HOÀN TẤT
	`NO_SHOW`	Persistent	Customer failed to attend within grace period (mutually exclusive with Completed).	VẮNG MẶT
Spa Confirmation (Protocol)	`PENDING`	Persistent	Dispatched at T-2h; awaiting customer acknowledgement.	CHỜ XÁC NHẬN
	`CONFIRMED`	Persistent	Customer clicked affirmative confirmation button.	ĐÃ XÁC NHẬN
	`DECLINED`	Persistent	Customer declined attendance or requested reschedule.	TỪ CHỐI
	`EXPIRED`	Persistent	Confirmation window lapsed at T-60m (Does NOT equal No-Show).	HẾT HẠN XN
Dining Order (`Meal_Order`)	`PENDING`	Persistent	Meal selected by customer; queued in kitchen prep system.	CHỜ XỬ LÝ
	`PREPARING`	Persistent	Chef commenced culinary cooking and ingredient assembly.	ĐANG CHẾ BIẾN
	`READY`	Persistent	Dish plated, inspected against allergens, ready for service/delivery.	ĐÃ SẴN SÀNG
	`DELIVERED`	Persistent	Delivered to restaurant table or customer sanctuary villa.	ĐÃ BÀN GIAO
Customer Folio (`Stay_Folio`)	`OPEN`	Persistent	Active stay; accepting extra charges from Spa (M3) and Dining (M4).	ĐANG MỞ
	`SETTLED`	Persistent	All line items paid; zero balance; cleared for checkout.	ĐÃ QUYẾT TOÁN
	`CLOSED`	Persistent	Stay finalized; locked against further debit or credit postings.	ĐÃ ĐÓNG
Runtime Evaluation Results (Non-Persistent)	`CANDIDATE_UNAVAILABLE`	**Runtime Event**	No therapist + room combination open for requested window.	Error banner ➔ Redirect to Screen 27
	`ALLERGY_CONFLICT_DETECTED`	**Runtime Event**	Dish contains allergen listed in customer health profile.	Red warning badge; dish masked/disabled
	`BILLING_GUARD_BLOCKED`	**Runtime Event**	Attempted checkout while unpaid charges or active orders exist.	Screen 44 Blocker Panel with line-item detail

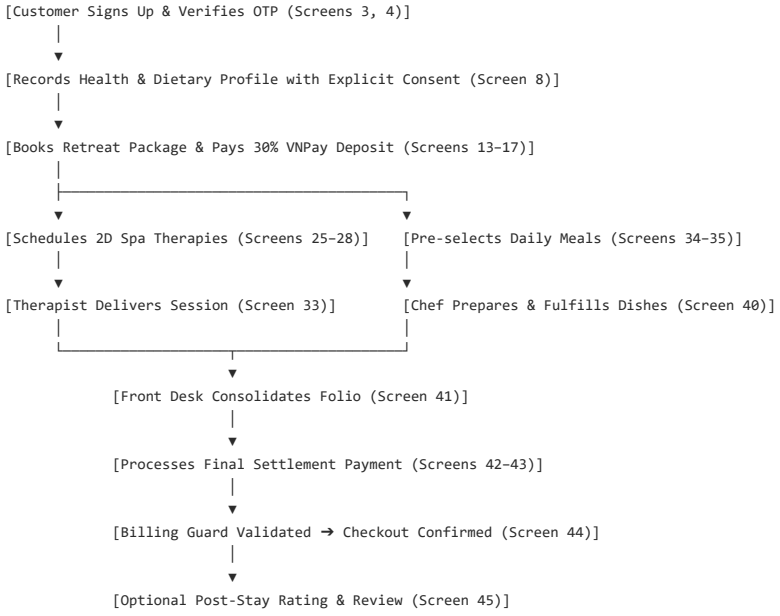
13. Cross-Module UI Dependencies & Data Flow

13.1 Cross-Module Data Handoff Architecture

No module in HSWRMS operates as a silo. Every guest experience and staff action relies on verified data published by preceding modules in the lifecycle:

ORIGINATING SOURCE	CONSUMING MODULE & SCREEN	TRANSFERRED DATA PAYLOAD	ENFORCED BUSINESS DEPENDENCY
M1: Health Profile (Screen 8)	**M3: Spa Session Detail** (Screen 33)	Musculoskeletal tension, mobility constraints, heat precautions.	Therapist must adapt treatment pressure and oils without viewing medical history.
M1: Health Profile (Screen 8)	**M4: Kitchen Prep List** (Screen 39)	Food allergies (Nuts, Shellfish, Gluten, Dairy), dietary regime.	Chef must segregate preparation utensils; allergens highlighted in bold red.
M2: Retreat Booking (Screen 15–17)	**M3: Spa Scheduling** (Screen 26)	Active stay dates, retreat package code, included therapy quota.	Spa appointments must fall strictly within retreat check-in and check-out dates.
M2: Retreat Booking (Screen 15–17)	**M4: Dietary Dining** (Screen 35)	Stay tenure, guest headcount, included daily meal entitlement.	Daily meal pre-selection available only for dates corresponding to booked stay.
M3: Extra Spa Booking (Screen 30)	**M5: Consolidated Folio** (Screen 41)	Service ID, treatment title, practitioner, price (VND), timestamp.	Extra therapy charges immediately append as pending debit line items on folio.
M4: A-la-carte Dining (Screen 36)	**M5: Consolidated Folio** (Screen 41)	Order ID, line items, quantity, total price (VND), delivery villa.	Culinary room-service charges automatically post to customer folio.
M2 + M3 + M4 (All Active Services)	**M5: Consolidated Invoice** (Screen 41–44)	Package balance + Spa extras + F&B extras + Statutory taxes.	Billing guard aggregates all debits; checkout blocked until folio reaches 0 balance.
M6: Business Params & Flags (Screen 50, 51)	**Modules 1 through 5** (All System Screens)	Deposit % (30%), refund windows, feature flags, RBAC rules.	Runtime behavior and feature availability dynamically governed across all screens.

13.2 Visual End-to-End Enterprise Flowchart

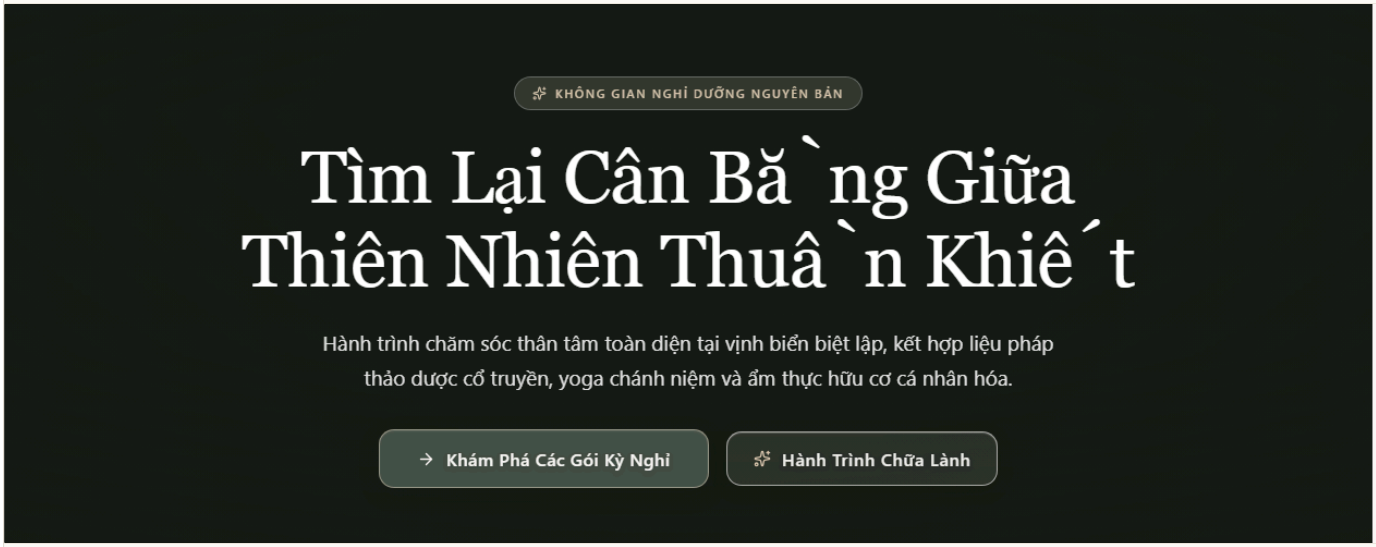


Visual UI/UX Walkthrough (Representative Screen Tour)

The following representative screens were captured directly from the live, running frontend prototype (React 19 / Vite preview server). Each figure documents the assigned screen number, primary actor, core operational purpose, key UI components, and verified interactive behavior.

Screen #1: Public Landing PUBLIC / SHARED

Public brand showcase presenting luxury wellness pillars, signature retreat packages, and multi-language access.

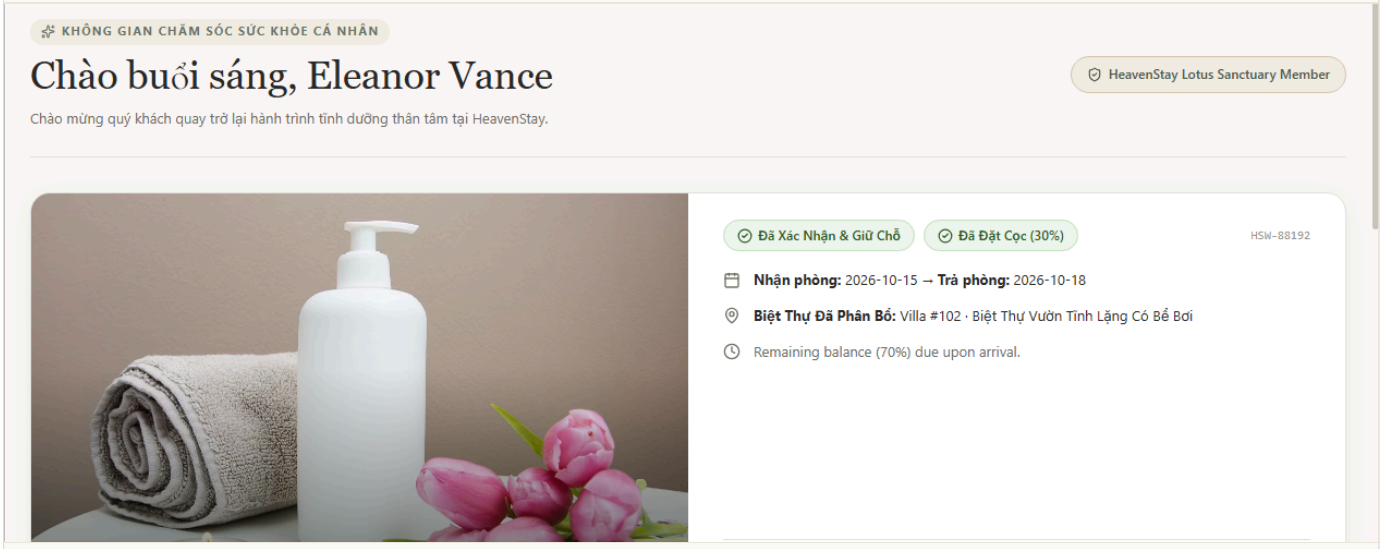


Key UI Components: Editorial hero typography, multi-language switcher dropdown (5 languages), featured retreat carousel, CTA buttons.

Important Interaction: Clicking "Khám Phá Các Gói Kỳ Nghỉ" navigates to Screen 13 catalog; changing language dynamically updates entire UI text.

Screen #2: Customer Dashboard CUSTOMER

Central authenticated hub for the in-house customer tracking active stay status, itinerary schedule, and wellness readiness.

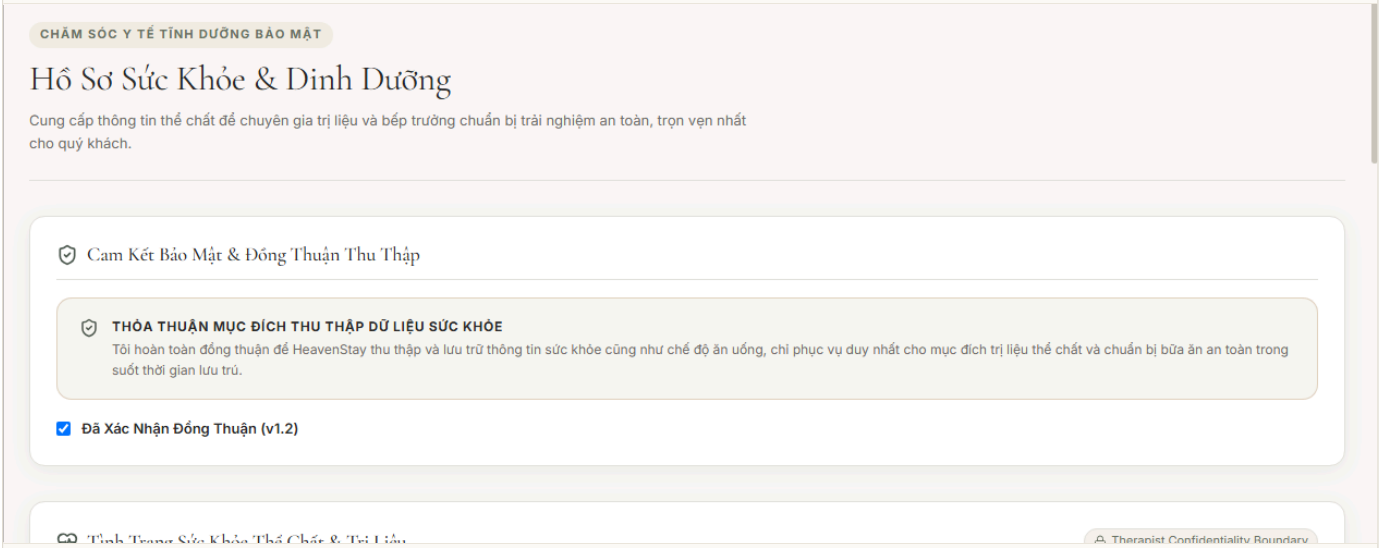


Key UI Components: Customer welcome banner, active retreat summary card (HSW-88192, Villa #102, 30% deposit paid), wellness readiness cards.

Important Interaction: Quick-action shortcuts navigate directly to Health Profile (Screen 8), Spa Booking (Screen 25), and Dining (Screen 34).

Screen #8: Health & Dietary Profile CUSTOMER

Confidential wellness intake form capturing physical precautions and food allergies under Vietnam PDPD consent standards.

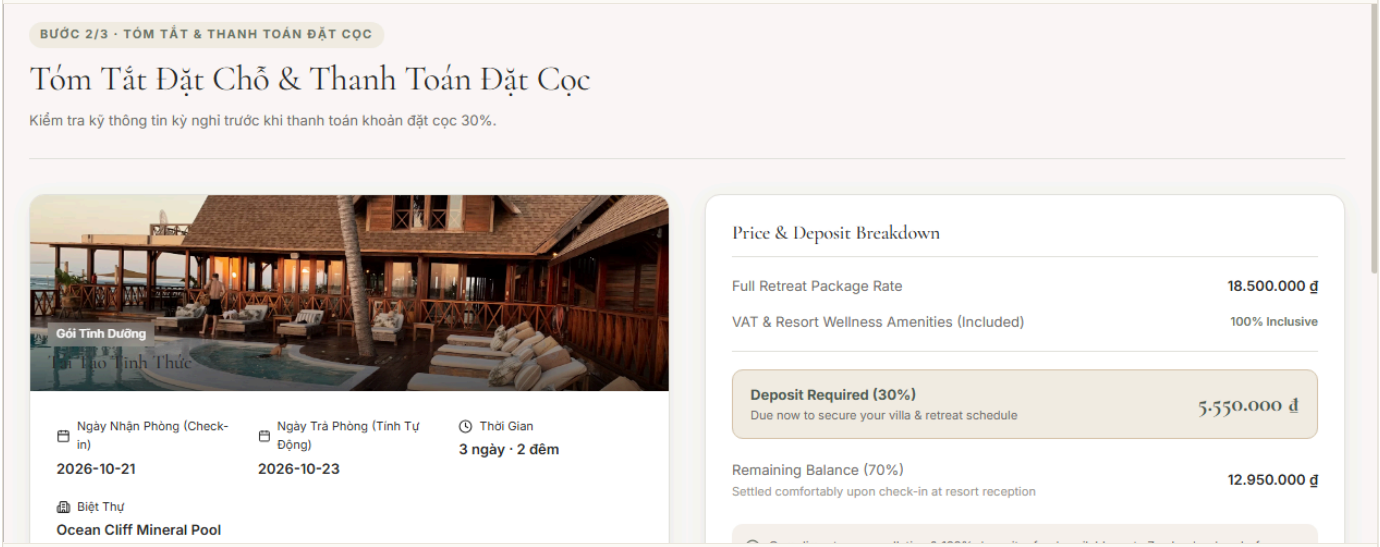


Key UI Components: Explicit unchecked-by-default consent checkbox, medical condition tags, mobility constraints, food allergy selector.

Important Interaction: Customer toggles health conditions; system enforces unchecked consent validation before saving profile to secure storage.

Screen #16: Booking Summary & Deposit CUSTOMER

Pre-payment booking verification calculating mandatory 30% retreat deposit and presenting cancellation refund policy.



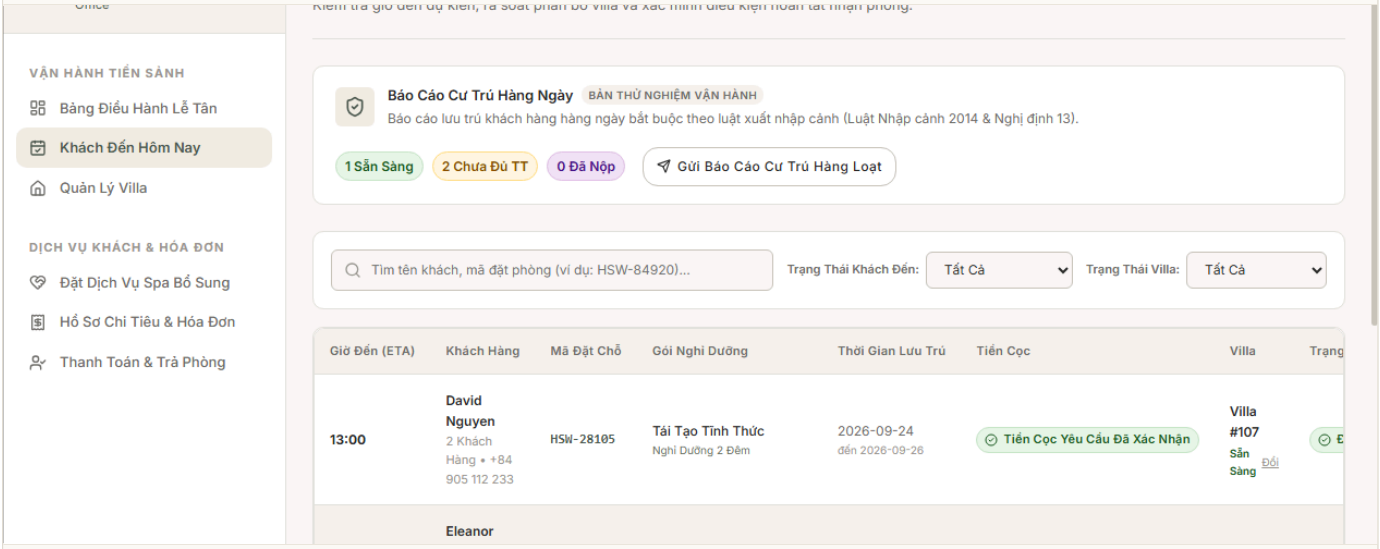
Key UI Components: Package itemization, dates, villa category, 30% deposit calculation, cancellation policy notice, VNPay payment CTA.

Important Interaction: Clicking "Thanh Toán Đặt Cọc (VNPay)" simulates redirect to VNPay gateway callback (Screen 17).

Screen #21: Daily Arrivals Queue

RECEPTIONIST

Front Desk daily arrival management displaying arriving guests, deposit verification, and statutory residence compliance.



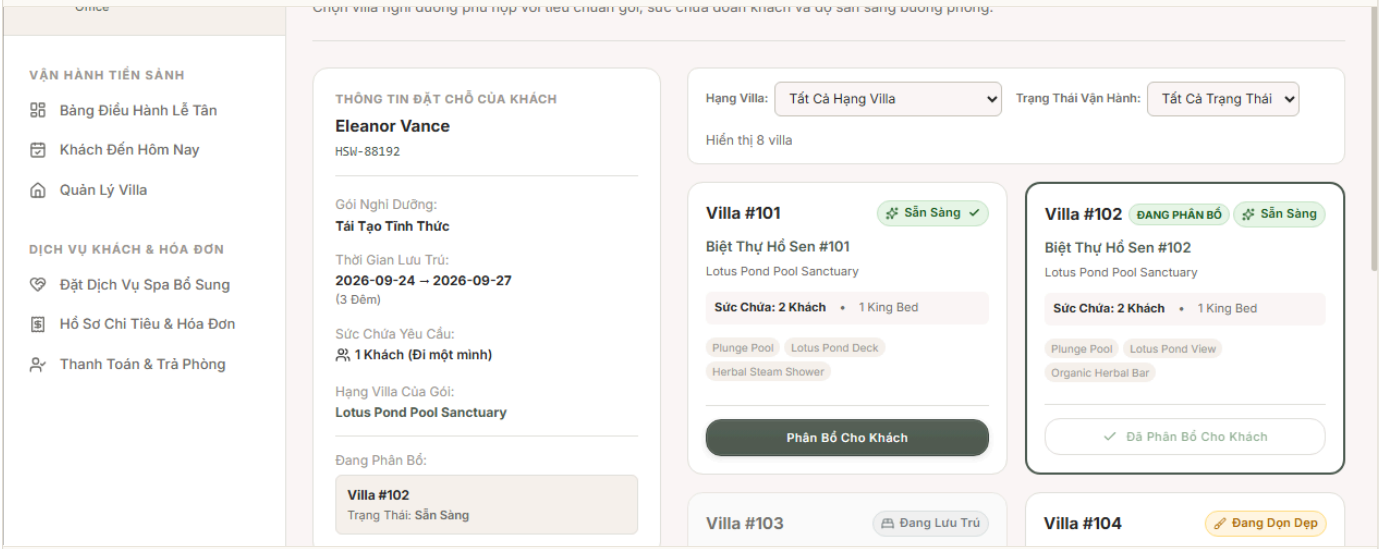
Key UI Components: Daily arrivals table, ETA indicators, booking reference, villa readiness tag, statutory Law on Residence banner.

Important Interaction: Receptionist clicks "Bắt Đầu Check-In" to initiate mandatory identification verification on Screen 22.

Screen #23: Physical Villa Assignment

RECEPTIONIST

Allocates a specific physical villa unit (e.g. Villa #102) matching the abstract villa category booked by the customer.

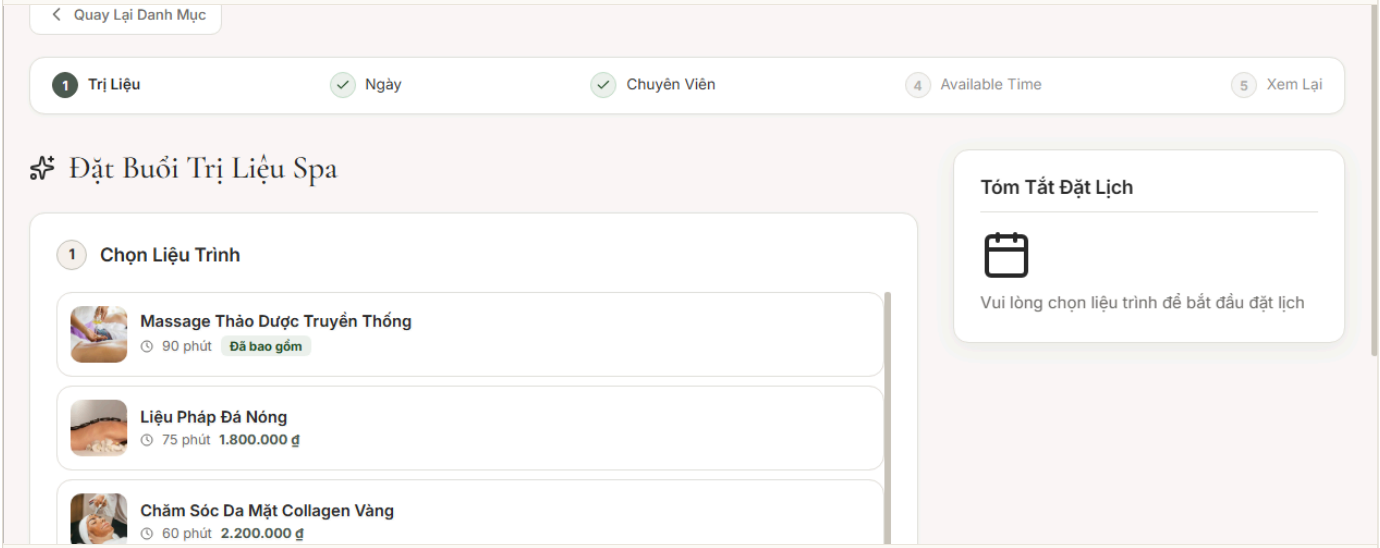


Key UI Components: Physical villa inventory grid, housekeeping status chips (Available, Cleaning, Maintenance), room view and floor tags.

Important Interaction: Receptionist verifies housekeeping readiness dot and clicks "Phân Bổ Villa" to bind physical room to guest stay.

Screen #26: Spa Session Booking CUSTOMER

Therapeutic appointment selection initiating system-driven atomic 2D resource matching (Therapist + Room).

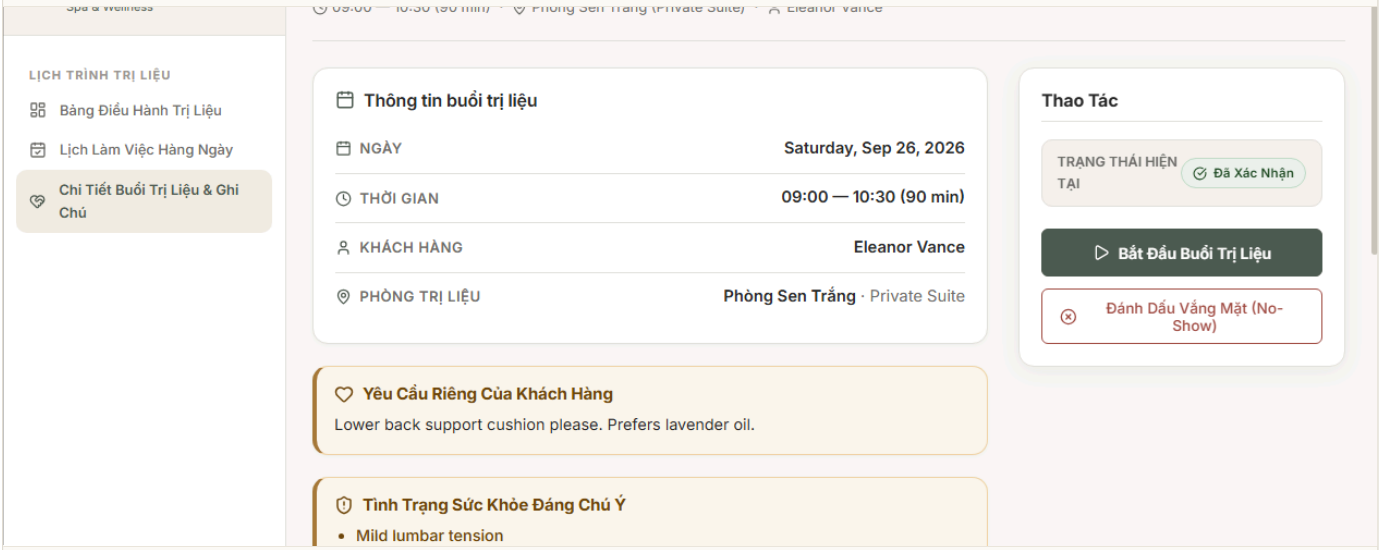


Key UI Components: 7-day date selector strip, available time slot grid, treatment duration pill, special request textarea, candidate preview tile.

Important Interaction: Selecting a date dynamically fetches open slots; choosing a slot locks both practitioner and treatment room.

Screen #33: Therapist Session Detail SPA THERAPIST / YOGA INSTRUCTOR

Clinical treatment delivery view projecting Treatment Safety Health Notes while strictly masking customer dietary and PII data.

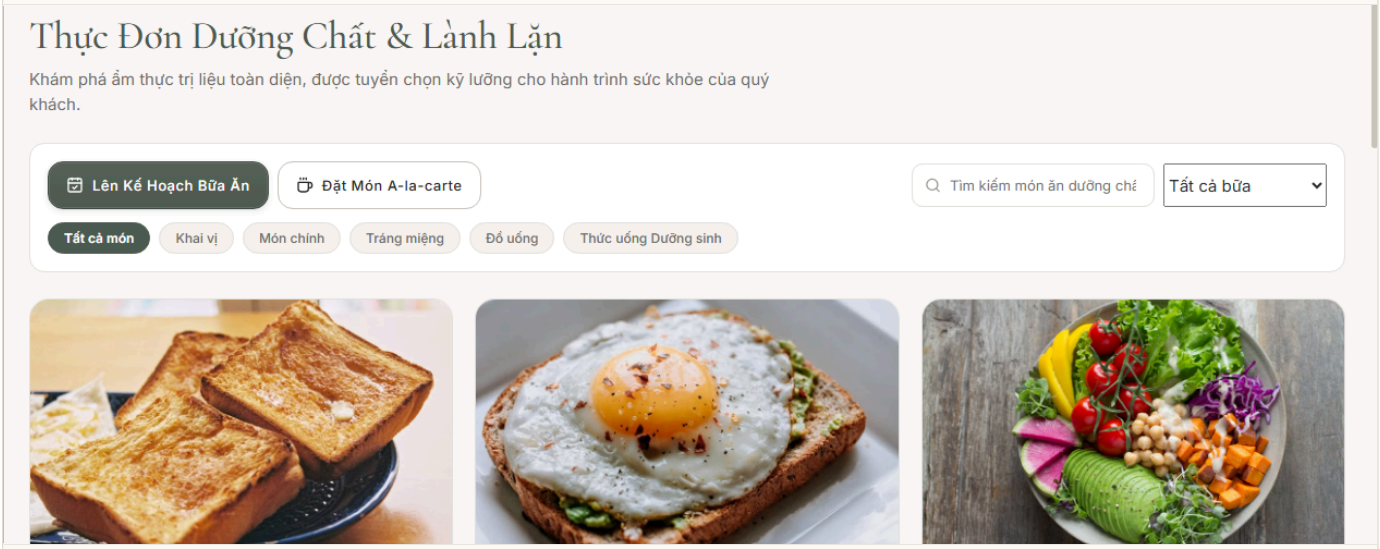


Key UI Components: Session time block, Treatment Safety Notes card (lumbar tension, hypertension), special requests, Completed & No-Show CTAs.

Important Interaction: Therapist reviews musculoskeletal precautions before starting session; marks Completed or No-Show upon conclusion.

Screen #34: Dietary Menu CUSTOMER

Organic dining catalog dynamically filtered according to customer allergy profile and nutritional goals.

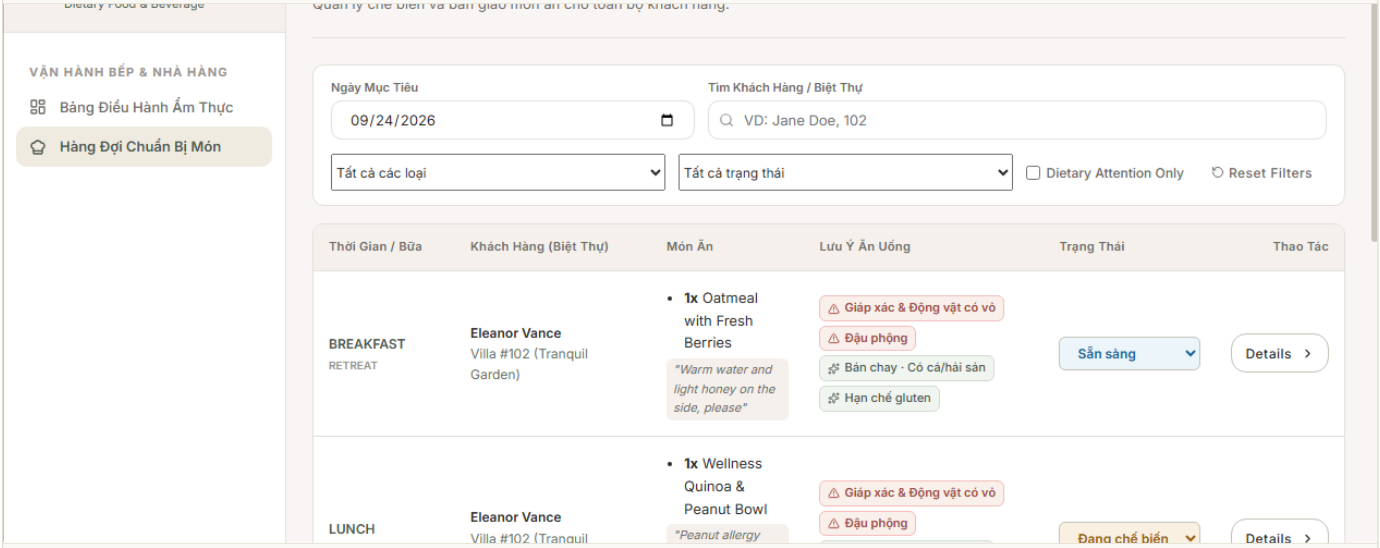


Key UI Components: Meal period tabs (Breakfast, Lunch, Dinner), dietary badges (Vegan, Pescatarian, Gluten-Conscious), allergen warning badges.

Important Interaction: Dishes containing allergens declared in Screen 8 are automatically flagged or masked from selection.

Screen #39: Daily Meal Prep List F&B STAFF / CHEF

Kitchen preparation command board aggregating meal portion requirements and highlighting critical culinary allergies.



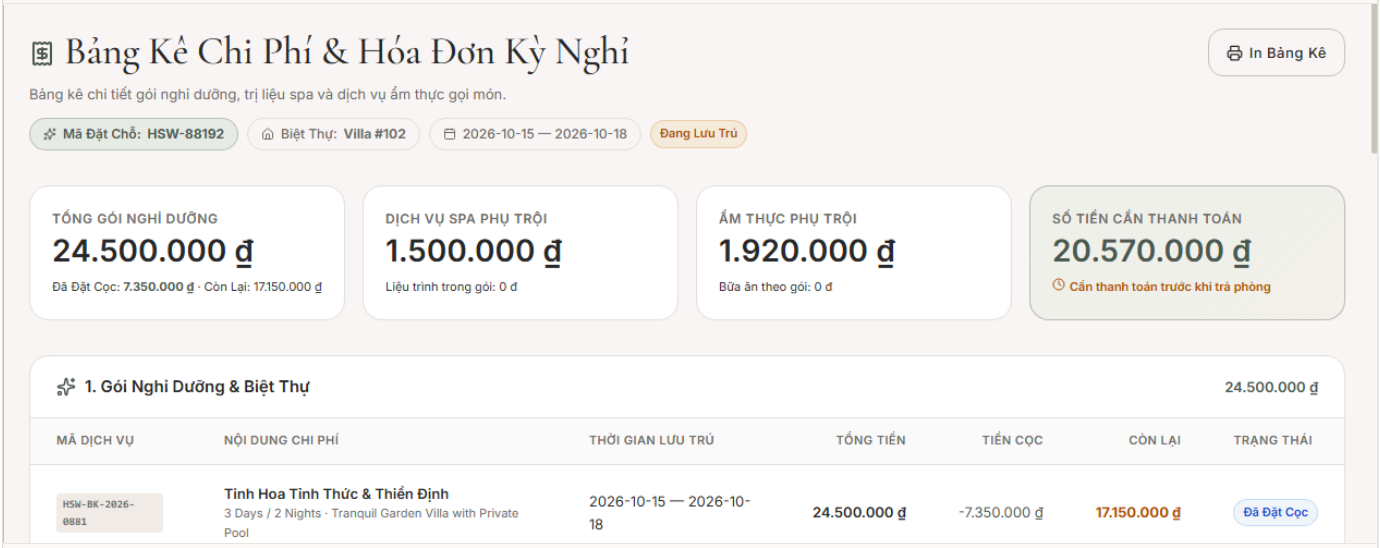
Key UI Components: Meal prep cards grouped by service period and villa delivery location, high-contrast red allergen alert badges (Nuts, Shellfish).

Important Interaction: Kitchen staff clicks into an order ticket (Screen 40) to transition meal fulfillment from Preparing to Ready.

Screen #41: Customer Folio & Consolidated Invoice

RECEPTIONIST / CUSTOMER

Itemized financial statement aggregating retreat package balance, extra spa treatments, and extra a-la-carte dining charges.



Key UI Components: Summary metric cards (Package Total, Spa Extras, Dining Extras, Outstanding Balance), itemized line-item audit table.

Important Interaction: Displays in read-only mode for Customer (`/customer/folio`) and operational settlement mode for Front Desk.

Screen #44: Checkout Confirmation & Departure Handover

RECEPTIONIST

Final departure handover enforcing the Billing Guard blocker and updating physical villa status to housekeeping cleaning.

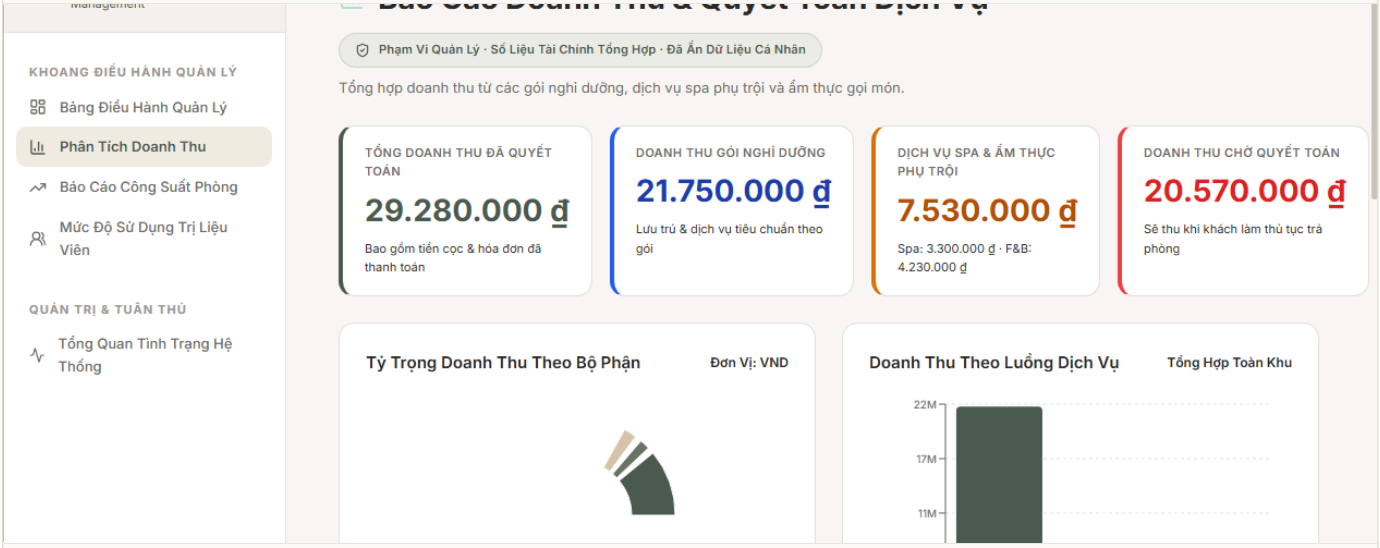


Key UI Components: Billing guard verification banner, checkout record details, villa turnover confirmation, review submission link.

Important Interaction: If unpaid charges remain, checkout is blocked; once settled, clicking finalize transitions villa to Vacant Needs Cleaning.

Screen #46: Manager Revenue Dashboard MANAGER

Executive financial reporting displaying revenue breakdowns across Package, Spa, and Dining without exposing guest PII.

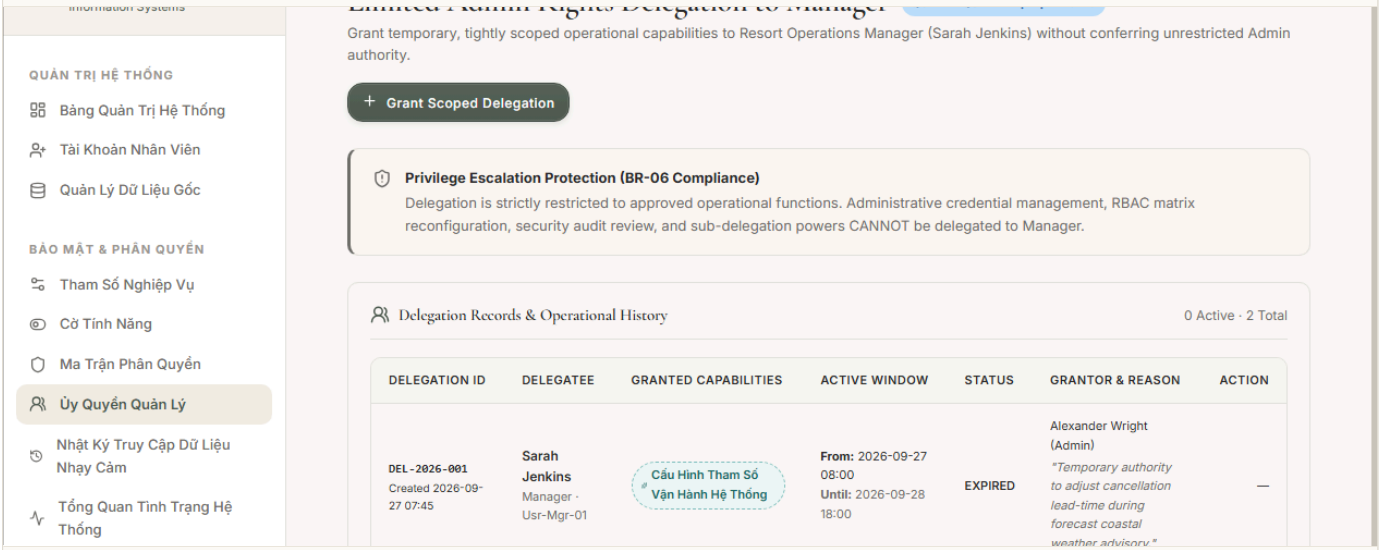


Key UI Components: Revenue summary cards, historical revenue bar chart, service category distribution donut chart, date range picker.

Important Interaction: Manager filters date horizon (Monthly, Quarterly) and triggers export of aggregated analytical reports.

Screen #53: Limited Admin Rights Delegation ADMIN

Governance console allowing Administrator to delegate tightly scoped operational capabilities to Manager with anti-escalation controls.



Key UI Components: Active delegation records table, capability selector, validity time window input, mandatory justification reason field.

Important Interaction: Admin creates temporary 24-hour delegation; system prevents delegation of master passwords or security audit logs.

14. Prototype vs. Approved Requirements (Gap Analysis)

Pursuant to the core evaluation rules of HSWRMS, discrepancies between the current React prototype and the latest approved SRS specifications are documented openly below with clear severity ratings and recommended remediation actions.

MODULE	SCREEN	CURRENT PROTOTYPE BEHAVIOR	APPROVED SRS / SPECIFICATION	SEVERITY	GAP & RECOMMENDED REMEDIATION
M3	Screen 26 (Spa Booking)	UI includes an optional `preferredTherapistId` dropdown allowing customer to choose a practitioner directly.	Customer selects only Treatment and Time Window; HSWRMS generates the complete candidate (Therapist + Room).	HIGH	**Manual Resource Selection vs. System Candidate:** Replace practitioner picker with pure time preference input; display practitioner and room only in the generated candidate card.
M3	Screen 28, 29 (Spa Reminders)	Mocked data displays a single `~1h reminder` timestamp (`reminderScheduledAt`).	5-tier reminder protocol: T-24h (info), T-2h (confirm prompt), T-75m (final), T-60m (deadline), T-15m (upcoming).	MEDIUM	**Simplified Reminder Timeline:** Update mock session card and reminder notification banner to reflect the multi-stage reminder milestones.
M4	Screen 34, 39 (Dietary Menu & Prep)	Menu items store static tags (`dietaryTags`, `allergenTags`); orders track item IDs directly.	Relational structure: `MENU_ITEM` ➔ `MENU_ITEM_INGREDIENT` ➔ `INGREDIENT` ➔ `INGREDIENT_BATCH` with batch math.	MEDIUM	**Missing Ingredient Model:** Add simulated ingredient quantities on the Kitchen Prep List (Screen 39) to demonstrate batch-level stock depletion.
M5	Screen 42 (Final Payment)	Payment console presents only the VNPay simulated gateway mode selector.	SRS BF-05 and UC22 explicitly describe both **Cash** (front-desk physical settlement) and **VNPay** payment channels.	MEDIUM	**Cash Settlement Omission:** Introduce a payment method toggle on Screen 42 allowing Receptionist to record a physical cash receipt.
M1	General / App Shell	Some localized strings in footer and legacy notes referenced "Guest" or "Khách Nghi".	`AGENTS.md` mandates: *"Use the term Customer consistently instead of Guest."*	LOW	**Terminology Standardization:** Global string audit performed; ensure all future translations strictly use "Customer" / "Khách Hàng".
M6	Screen 9 vs Admin	Screen 9 allows Customer to request health data deletion; Admin side soft-delete is simulated in data layer.	SRS Module 6 diagram specifies an Admin task: *"Manage Data Privacy & Deletion (Execute Soft-Delete)"*.	LOW	**Missing Admin Deletion Queue:** Add a dedicated "Privacy Deletion Requests" tab in Screen 54 (Audit Log) or Screen 10.

15. UI/UX Quality & Usability Review

An independent design and human-factors audit was conducted across the 56-screen frontend prototype against 14 established usability criteria:

EVALUATION CRITERION	RATING	OBSERVED STRENGTHS	IDENTIFIED RISKS & RECOMMENDATIONS
Navigation Clarity	GOOD	Hierarchical breadcrumbs, persistent sidebars for staff, and unified top navigation for customers provide immediate orientation.	Deep nested paths (e.g. villa assignment) rely on breadcrumb clicks; ensure back buttons exist on every step.
Role Consistency	GOOD	Distinct visual shells (Customer topbar vs. Staff 260px sidebar) prevent accidental persona confusion.	RoleSwitcher dev tool must be strictly stripped or disabled in production deployments via feature flags.
Information Hierarchy	GOOD	Serif headers paired with neutral sans-serif data grids create an elegant, scannable reading rhythm.	Some analytics cards contain high metric density; maintain adequate white space on smaller laptop viewports.
Visual Consistency	GOOD	Strict enforcement of tokens (`tokens.css`); universal 18px card radii and subdued earth tones.	Ensure third-party icons from Lucide maintain uniform 1.5px stroke weight throughout.
Interactive Feedback	GOOD	Immediate toast notifications for role switching, form saves, and cart operations; smooth hover micro-lifts.	Simulated async network delays (e.g. payment processing) should consistently display loading skeleton spinners.
Form Validation	ACCEPTABLE	Mandatory unchecked consent checkbox on health profile; password strength validation on registration.	Add inline field-level error messages alongside toast alerts for multi-field forms like residence declarations.
Empty States	GOOD	Dedicated empty state illustrations and helpful discovery CTAs when bookings or spa lists are blank.	Maintain localized Vietnamese copy across all empty state callouts.
Error Handling	GOOD	Billing guard blocker panel on Screen 44 explicitly details why checkout cannot proceed.	Ensure simulated payment failure gives clear recovery actions (e.g. retry with another card).
Status Visibility	GOOD	Color-coded status chips (Confirmed, Cleaning, Occupied) make operational states instantly recognizable.	Maintain high contrast between badge backgrounds and text for WCAG AA compliance.
Accessibility (a11y)	ACCEPTABLE	Semantic HTML headings, button elements, and clear label-to-input associations across major forms.	Audit low-contrast muted gray text (`#706E6B`) against off-white background to ensure 4.5:1 ratio.
Responsive Layout	ACCEPTABLE	Fluid container system with mobile-friendly flex wrapping on customer and public pages.	Staff operational data tables (e.g. arrivals, audit log) require horizontal scroll containers on tablets.
Data Privacy Guard	GOOD	Therapist view masks food and PII; Chef view masks clinical health; Receptionist sees zero medical data.	Reiterate to evaluators that backend API projections must enforce the same filtering as the front-end views.
Action Clarity	GOOD	Action verbs used on buttons ("Thanh Toán Đặt Cọc", "Hoàn Tất Trả Phòng", "Phân Bổ Villa").	Keep primary CTA visually dominant while secondary actions remain ghost or secondary styles.
Destructive Confirmations	GOOD	Modal dialog confirmations required for session cancellation, villa maintenance lock, and data deletion.	Maintain explicit typed confirmation phrases for irreversible actions like health data deletion.

16. Demo Walkthrough Script (10–15 Minutes)

This structured demonstration script is designed for an end-to-end presentation during the ISP392 project defense. It demonstrates the complete 6-module lifecycle across 4 distinct personas in approximately 12 minutes.

STEP	PERSONA & ROLE	SCREEN VISITED	DEMONSTRATED ACTION	KEY TALKING POINT & BUSINESS RULE FOR EXAMINERS
1	PUBLIC VISITOR	Screen #1 (Landing Page)	Browse editorial landing, switch language (VI ➔ EN), explore retreat packages.	Quiet luxury aesthetic inspired by Amanoi. Explain that retreat packages combine accommodation, dining, and daily therapies as a unified product.
2	CUSTOMER	Screen #8 (Health Profile)	Open health profile; review medical and allergy tags; observe unchecked consent box.	**Vietnam PDPD Compliance:** Health data requires explicit, non-pre-checked consent. Medical notes are segregated from general profile data.
3	CUSTOMER	Screen #15 ➔ 17 (Retreat Booking)	Select stay dates and villa category; review 30% deposit; simulate VNPay success.	Customer books an abstract **Villa Category**, not a physical room number. Deposit confirmation generates a locked retreat booking.
4	RECEPTIONIST	Screen #21 ➔ 23 (Arrivals & Check-In)	Switch to Receptionist; open Daily Arrivals; verify guest ID; assign physical Villa #102.	**Law on Residence 2020:** Mandatory statutory declaration. Explain physical villa assignment decoupling and villa status transition to *Occupied*.
5	CUSTOMER	Screen #26 ➔ 28 (Spa Scheduling)	Switch to Customer; book included herbal therapy; review candidate time slot; confirm.	**Atomic 2D Lock:** System simultaneously validates both qualified therapist and treatment room. Reminders follow T-24h to T-15m protocol.
6	CUSTOMER	Screen #34 ➔ 36 (Dining & Extras)	Pre-select daily dinner course; order an extra artisanal wellness tea charged to folio.	Dietary menu filtered by customer allergy profile. Extra a-la-carte item automatically posts pending debit to the Customer Folio (`M4` ➔ `M5`).
7	THERAPIST	Screen #32 ➔ 33 (Therapist Schedule)	Switch to Therapist; open daily agenda; view session details; note health precautions.	**Data Minimization:** Therapist sees only muscular/joint notes ("Lumbar tension"). Full medical history, dietary info, and payment records are masked.
8	F&B STAFF / CHEF	Screen #39 ➔ 40 (Kitchen Prep List)	Switch to Chef; review morning prep board; inspect allergen warnings; mark dish Ready.	Chef sees critical allergen warnings (Nuts/Gluten) prominently in red while medical records are completely hidden. Status transitions to Ready.
9	RECEPTIONIST	Screen #41 ➔ 44 (Folio & Checkout)	Open Folio; show aggregated line items (Package + Spa + Tea); pay balance; confirm checkout.	**Payment Success != Checkout Complete:** Explain billing guard blocking. Checkout updates villa to *Vacant Needs Cleaning* and releases booking.
10	MANAGER	Screen #46, 55 (Revenue & Dashboard)	Switch to Manager; review revenue breakdowns, villa occupancy trends, therapist hours.	**Least-Privilege Role:** Manager has read-only analytical visibility over operations without possessing administrator credentials.
11	ADMINISTRATOR	Screen #51, 53, 54 (Governance & Audit)	Switch to Admin; show feature flag toggles, delegated permissions, and sensitive audit log.	Demonstrate anti-escalation controls on delegation (Screen 53) and immutable logging of all health record views (Screen 54).

17. Mock Viva: 20+ Challenging Defense Questions

Anticipated technical and architectural examination questions for the ISP392 project defense, accompanied by examiner intent and model answers:

Q1: Why does the Customer select a "Villa Category" online instead of picking a specific physical villa room number? M2 Architecture

Tests candidate's understanding of real-world resort yield management and housekeeping constraints.

Model Answer: In luxury wellness hospitality, assigning specific physical rooms weeks in advance creates catastrophic operational bottlenecks if a room requires unscheduled maintenance or a previous guest extends their stay. Reserving a Villa Category guarantees the guest's booked tier while granting Front Desk receptionists the operational flexibility to allocate inspected, ready physical villas on arrival date (Screen 23).

Q2: Why does the Spa scheduling module require an "Atomic 2D Lock" rather than letting guests pick any therapist? M3 Concurrency

Tests whether the team understands resource contention in multi-asset scheduling.

Model Answer: A spa therapy requires both an accredited practitioner (Therapist) and a specialized physical environment (Treatment Room with hydrotherapy baths or massage tables). If the system reserved a therapist without reserving a room, two therapists could be scheduled into the same room simultaneously. The atomic 2D lock reserves both resources simultaneously in a single transaction, preventing deadlocks and resource conflicts.

Q3: Why is "Payment Success" decoupled from "Checkout Completion" in Module 5? M5 Domain Modeling

Evaluates understanding of hospitality operational lifecycles vs. payment processing.

Model Answer: Payment success is purely a financial settlement transaction confirming funds transfer. Checkout completion is an operational resort lifecycle event. Finalizing checkout requires verifying that no pending laundry or minibar charges exist, closing the customer folio, updating the guest stay to 'Checked-Out', and triggering housekeeping by changing the physical villa status to 'Vacant Needs Cleaning'.

Q4: Why must the consent checkbox on Screen 8 be explicitly unchecked by default? M1 Legal Compliance

Verifies compliance with personal data protection legislation.

Model Answer: Under Decree No. 13/2023/NĐ-CP on Personal Data Protection (PDPD) and international GDPR standards, processing sensitive personal data (health and medical history) requires clear, affirmative, and unambiguous consent. Pre-ticked checkboxes are legally invalid because silence or inactivity does not constitute consent.

Q5: How does the system enforce data minimization between the Therapist and the Chef? Data Privacy

Tests understanding of principle of least privilege applied to health data.

Model Answer: Rather than exposing the full customer profile object, HSWRMS creates role-adapted data projections. The TherapistSafeWellnessView projects only musculoskeletal and pressure precautions, omitting dietary info and identity details. Conversely, the FnbDietaryProjection projects only food allergies and dietary goals, strictly concealing musculoskeletal injuries and clinical history.

Q6: Why is the Manager role not granted full Administrator privileges? M6 Security Architecture

Evaluates adherence to separation of duties and least-privilege principles.

Model Answer: Separation of Duties is a foundational security control. A resort operations manager oversees day-to-day hospitality, occupancy, and staff schedules. Giving the manager administrative rights would allow them to alter security audit logs, modify system feature flags, or change staff access credentials without oversight. Administrative powers remain isolated with the System Administrator.

Q7: What is the purpose of the Limited Admin Rights Delegation feature (Screen 53)? M6 Governance

Tests ability to handle real-world operational exceptions securely.

Model Answer: During operational emergencies (e.g., weather advisories requiring changes to cancellation lead times), a manager may need to adjust business parameters when the administrator is unavailable. Screen 53 enables the admin to delegate tightly scoped capabilities for a limited time window with mandatory justification logging, preventing permanent privilege creep.

Q8: Why does the system support Screen 9 (Health Data Deletion Request)? M1 Privacy Rights

Tests understanding of data lifecycle management and guest privacy rights.

Model Answer: Guests frequently share sensitive personal health conditions solely for the duration of a retreat. Following checkout, they have the right to request deletion or anonymization of this health profile. Screen 9 captures this request while retaining financial transaction records required by statutory tax and accounting laws.

Q9: Why can't a customer complete checkout online from their own mobile phone without receptionist verification? M5 Front Desk Workflow

Validates realistic understanding of physical resort operations.

Model Answer: Luxury villa resorts require physical room key return, minibar consumption reconciliation, and inspection of sanctuary facilities. Furthermore, Vietnamese residence reporting laws require recording the departure timestamp at the front desk. The customer portal provides read-only folio access, but final checkout remains controlled by the Receptionist.

Q10: In Module 3, why does an "EXPIRED" confirmation status not equal a "NO_SHOW"? — M3 State Modeling

Tests precision of state machine boundaries.

Model Answer: Confirmation status tracks customer engagement at T-60 minutes before a session. If the customer does not confirm, the status expires, allowing the resort to contact the villa or release the therapist. A No-Show occurs only after the session start time has passed and the customer failed to attend. Treating unconfirmed sessions as No-Shows would unfairly penalize guests.

Q11: Why is Screen 41 implemented as a shared, role-adapted component instead of two separate screens? — Screen Inventory Discipline

Checks consistency with project screen list constraints.

Model Answer: The baseline screen list contains exactly 56 screens, with Screen 41 titled "Customer Folio & Consolidated Invoice". Implementing separate customer and receptionist screens would arbitrarily expand the screen count. Instead, `FolioInvoicePage` adapts its UI based on the active role: read-only for Customer, and operational settlement controls for Receptionist.

Q12: How does the system prevent race conditions during spa appointment scheduling? — M3 Concurrency

Tests backend concurrency concepts in the UI architecture.

Model Answer: When a candidate slot is generated, HSWRMS places a temporary reservation lock on both the therapist and room. Before final confirmation on Screen 28, the system revalidates the lock. If another session locked the resources in the interim, the transaction fails safely and redirects the customer to Screen 27 (Alternative Slots).

Q13: Why does the Kitchen Prep List (Screen 39) aggregate orders by service period rather than by customer order time? — M4 Culinary Operations

Tests domain knowledge of culinary kitchen workflows.

Model Answer: Commercial resort kitchens prep mise-en-place in batches by service period (Breakfast, Lunch, Dinner). Displaying prep lists chronologically by order time would cause kitchen staff to cook individual dishes inefficiently. Aggregating by meal period allows the kitchen to prepare total portions while sorting by delivery villa.

Q14: What prevents unauthorized staff from modifying feature flags or business parameters? — M6 Security

Tests RBAC implementation and privilege escalation guards.

Model Answer: In the frontend prototype, `RoleGuard` validates required roles and capabilities before rendering routes. On the backend, API endpoints enforce token-based claims verification, and protected core flags (e.g. security audit logging) cannot be modified even under delegated manager access.

Q15: Why is the post-stay Review & Rating (Screen 45) optional rather than a mandatory step of checkout? — M5 Customer Experience

Validates customer journey design logic.

Model Answer: Forcing customers to complete a review before departing creates high friction and delays check-out queues at the front desk. Per SRS UC23, Review & Rating is optional and accessible either via a post-checkout link on Screen 44 or an emailed feedback link sent after departure.

Q16: Why is the Police Residency Declaration report accessible from Screen 56? — Legal Compliance

Tests understanding of statutory Vietnamese hospitality laws.

Model Answer: Under the Law on Entry, Exit, Transit, and Residence of Foreigners in Vietnam (Law No. 47/2014/QH13) and Decree 144/2021/NĐ-CP, hospitality establishments must report daily guest residence data to local police authorities. Placing the export affordance in Screen 56 gives management a direct compliance reporting tool.

Q17: Why does the system use Vanilla CSS tokens instead of Tailwind CSS? — Architecture & Styling

Tests rationale for technical design system choices.

Model Answer: The project specification (`HSWRMS_DESIGN.md §17`) defines a bespoke luxury design token system. Vanilla CSS custom properties provide native, zero-runtime variable scoping, perfect theme encapsulation, seamless print media style overrides for PDF generation, and complete freedom from utility framework bloat.

Q18: How does the prototype handle localization across 5 languages? — i18n Architecture

Tests internationalization design in the codebase.

Model Answer: Localization is managed via `react-i18next` with structured JSON locale files for Vietnamese, English, French, Japanese, and Simplified Chinese. Complex domain entities (e.g. treatments, dishes, packages) utilize localized string objects (`{ vi, en, ja... }`) resolved dynamically via custom localization helper functions.

Q19: If an external payment gateway goes offline, how does HSWRMS maintain operational continuity? — Fault Tolerance

Tests resilience and offline fallback design.

Model Answer: At Front Desk checkout, Receptionist can switch payment method from VNPay to Cash or Offline Card Terminal (Screen 42). The receptionist records the authorization code or physical receipt number, allowing departure handover to proceed without stranding departing guests.

Q20: Why does the prototype use a client-side RoleSwitcher toolbar? Demo Tooling

Evaluates candidate's understanding of prototype evaluation ergonomics.

Model Answer: During project evaluation and defense, examiners must inspect how six different personas experience the system without constantly logging out and retyping credentials. The RoleSwitcher simulates persona switching instantly, updates context state, and navigates to the role's default workspace with visual feedback.

18. Open Architectural Items & Known Limitations

This section catalogues technical and business decisions that remain open or pending final stakeholder resolution:

OPEN ITEM	CURRENT KNOWN REQUIREMENT	WHAT REMAINS UNDECIDED	PROTOTYPE SIMULATION STRATEGY
M3/M4 ➔ M5 Folio Posting	Extra spa and dining orders must post debit lines to the customer stay folio.	Whether folio posting uses direct synchronous REST mutation, domain event pub/sub, or message queue.	Simulated via shared React context (`FolioContext`) observing active bookings and appending line items.
M3 Notification Channel	Reminders must trigger at T-24h, T-2h, T-75m, T-60m, and T-15m.	Mix of Web Push, SMS (e.g. Twilio/eSMS), and SendGrid Email in production.	Simulated via mock reminder timestamps on session records and in-app feedback toasts.
Cash vs. VNPay Ownership	Checkout payment supports VNPay gateway and front-desk cash.	Exact accounting journal entry structure for physical cash drawer balance reconciliation.	Current UI models VNPay gateway flow; Cash toggle flagged in Gap Analysis for backend phase.
M4 In-Villa Delivery	Dining orders can be delivered to customer sanctuary villas.	Whether delivery involves dedicated golf-cart dispatch tracking or standard room-service delivery.	Screen 36 and 40 record target villa destination; live GPS tracking excluded from current baseline.
M3 IN_PROGRESS State	Therapist can mark session Started/In-Progress.	Whether timer tracking is strictly required for clinical billing disputes.	Screen 33 includes "Bắt Đầu Buổi Trị Liệu" button transitioning status to `in_progress`.
M4 Ingredient Quantity Model	Menu items link to ingredient batches for nutritional and stock math.	Detailed database schema for ingredient waste percentage and seasonal supplier cost fluctuations.	Simulated via static dietary tags and dish descriptions; full batch math scheduled for backend.

19. Technical Frontend Summary

19.1 Technical Architecture Specification

Core Runtime & Libraries

- **Framework:** React 19 (`^19.2.8`) with TypeScript (`~6.0.2`)
- **Build Tool:** Vite 8 (`^8.3.0`) with native ESM hot reload
- **Routing:** React Router v7 (`^7.18.4`) with nested layouts & guards
- **Internationalization:** `i18next` (`^26.4.2`) & `react-i18next` (5 locales)
- **Typography & Icons:** Google Fonts (Cormorant Garamond + Inter) & `lucide-react` (`^1.48.0`)
- **Data Visualization:** `recharts` (`^3.10.1`) for revenue & occupancy charts

Source Topology & Project Layout

- `src/app/`: Top-level router, role guards, layout definitions
- `src/components/`: Common, navigation, customer, dining, and operations UI components
- `src/context/`: Domain state stores (Customer, Operations, Spa, Dining, Folio, Governance)
- `src/data/`: Realistic mock catalogs (Retreats, Spa, Dining, Staff, Governance)
- `src/i18n/`: Multi-language dictionaries (vi, en, fr, ja, zh-CN)
- `src/pages/`: 56 screen page components grouped cleanly by actor domain
- `src/styles/`: Design tokens, layout containers, and domain-specific CSS

19.2 Build Verification & Compilation Baseline

The frontend codebase compiles cleanly to static web assets using `npm run build` (`tsc -b && vite build`). The entire 56-screen application compiles in approximately **1.24 seconds**, producing clean production bundles with zero TypeScript type errors.

20. Final Assessment & Defense Readiness

20.1 Current Strengths of the Web UI Demo

- **100% Screen Inventory Completeness:** All 56 official baseline screens from `Screen_List.md` are fully implemented and navigable without missing placeholder links.
- **Authentic Quiet-Luxury Aesthetic:** Strict compliance with `HSWRMS_DESIGN.md` tokens creates an immediate impression of an ultra-high-end wellness retreat.
- **Strong Role & Data Isolation:** Rigorous separation of staff views guarantees sensitive health and dietary data minimization.
- **Interactive Domain State:** Fully simulated bookings, atomic spa scheduling, kitchen prep tickets, and consolidated folio billing allow fluid demonstration.

20.2 Presentation Readiness Categorization

Presentation-Ready Screens (Ready for Viva)

Screens 1–8, 10–25, 28–33, 35–41, 43–50, 52–56 are completely aligned with specifications, visually polished, and fully ready for presentation scrutiny.

Screens Requiring Business Synchronization

Screen 26 (Spa Booking candidate generator), Screen 39 (Ingredient batch math), and Screen 42 (Cash toggle addition) should be addressed prior to final code freeze.

20.3 Recommended Final QA Checklist Prior to Defense

- [x] Verify dev preview server runs cleanly on local port without console warnings.
- [x] Confirm all 56 routes map to valid page components without 404 fallbacks.
- [x] Ensure RoleSwitcher persona toggle correctly transitions contexts across all 6 roles.
- [x] Test language switcher across English and Vietnamese to ensure no missing translation keys.
- [x] Verify billing guard on Screen 44 blocks departure when folio has outstanding charges.
- [x] Confirm treatment-relevant health notes on Screen 33 conceal food allergies and PII.
- [x] Ensure presentation device resolution (1920×1080) renders all operational tables cleanly.